

HANDS-ON LAB WORKBOOK

Google Cloud Platform

Practical Cloud Exercises

A step-by-step lab manual for engineering students.
Build, connect, and verify real resources across the five core service families.

AUDIENCE

Engineering Students

DURATION

~2 Hours

FORMAT

5 Guided Labs

Lecture by **Imran Sayyed**



What You Need Before You Start

A short checklist to make sure your lab environment is ready.



GOOGLE ACCOUNT

A Google login with
Cloud Console access.



WEB BROWSER

Chrome or any modern
browser; no install
required.



BILLING / FREE TIER

A billing account or
free-trial credits
enabled.



A PROJECT

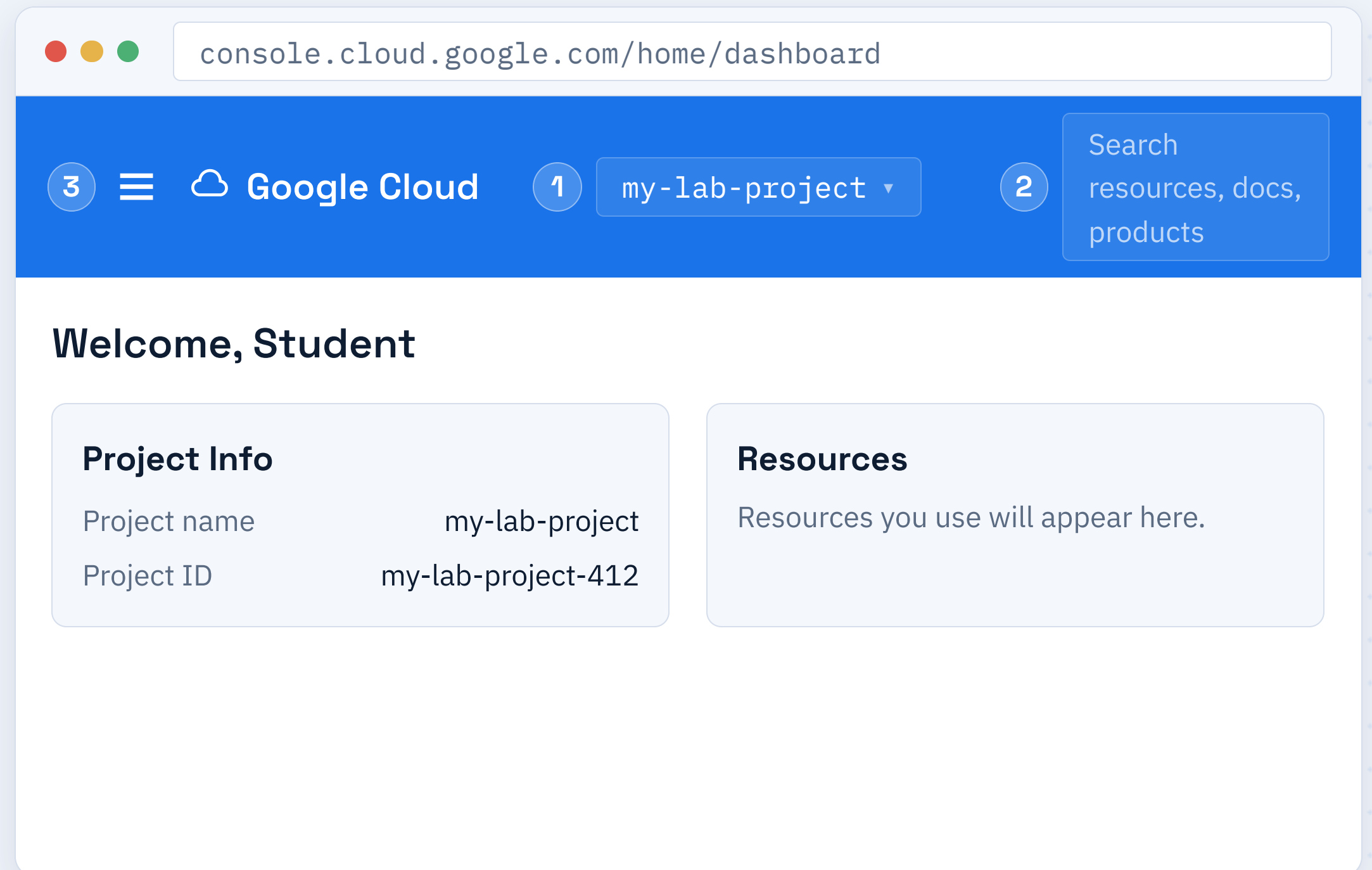
A dedicated project to
hold all lab resources.

Touring the Cloud Console

KEY AREAS

- 1 **Project picker** — top bar dropdown.
- 2 **Search bar** — jump to any service.
- 3 **Navigation menu** — the hamburger at top-left.

✓ **Expected result:** you can identify the project picker, search, and the navigation menu.

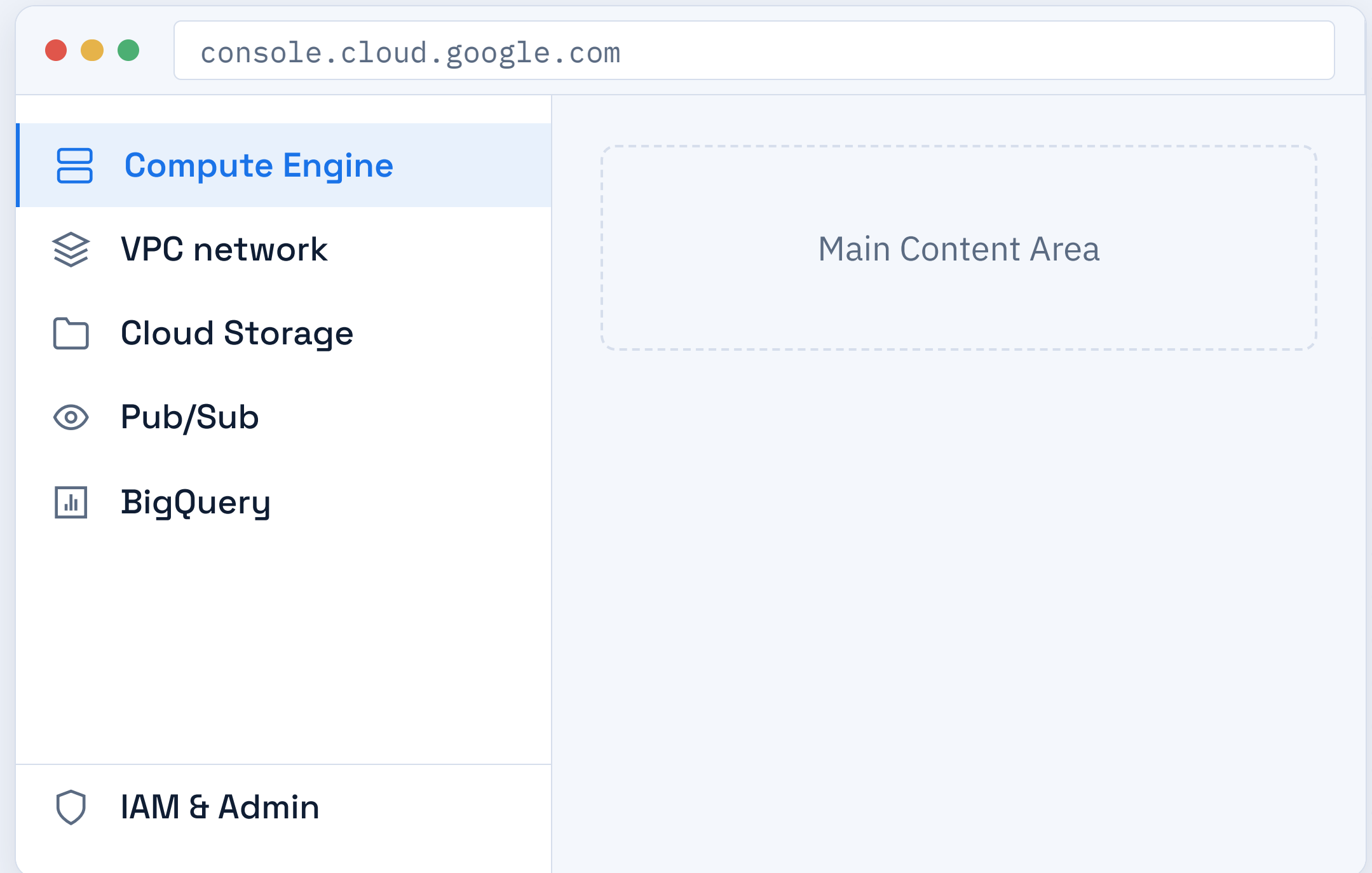


Finding Services in the Menu

STEPS

- 1 Click the **navigation menu** (≡) top-left.
- 2 Scroll to a product group such as **Compute** or **Storage**.
- 3 Pin frequently used services for quick access.

✓ **Expected result:** the navigation menu opens and lists product groups.



How Projects Organize Your Work

Every resource you create lives inside a project.



PROJECT

A container for resources, billing, and permissions.



PROJECT ID

A globally unique identifier you cannot change later.



ISOLATION

Resources in one project are separate from another.

Organization



Project



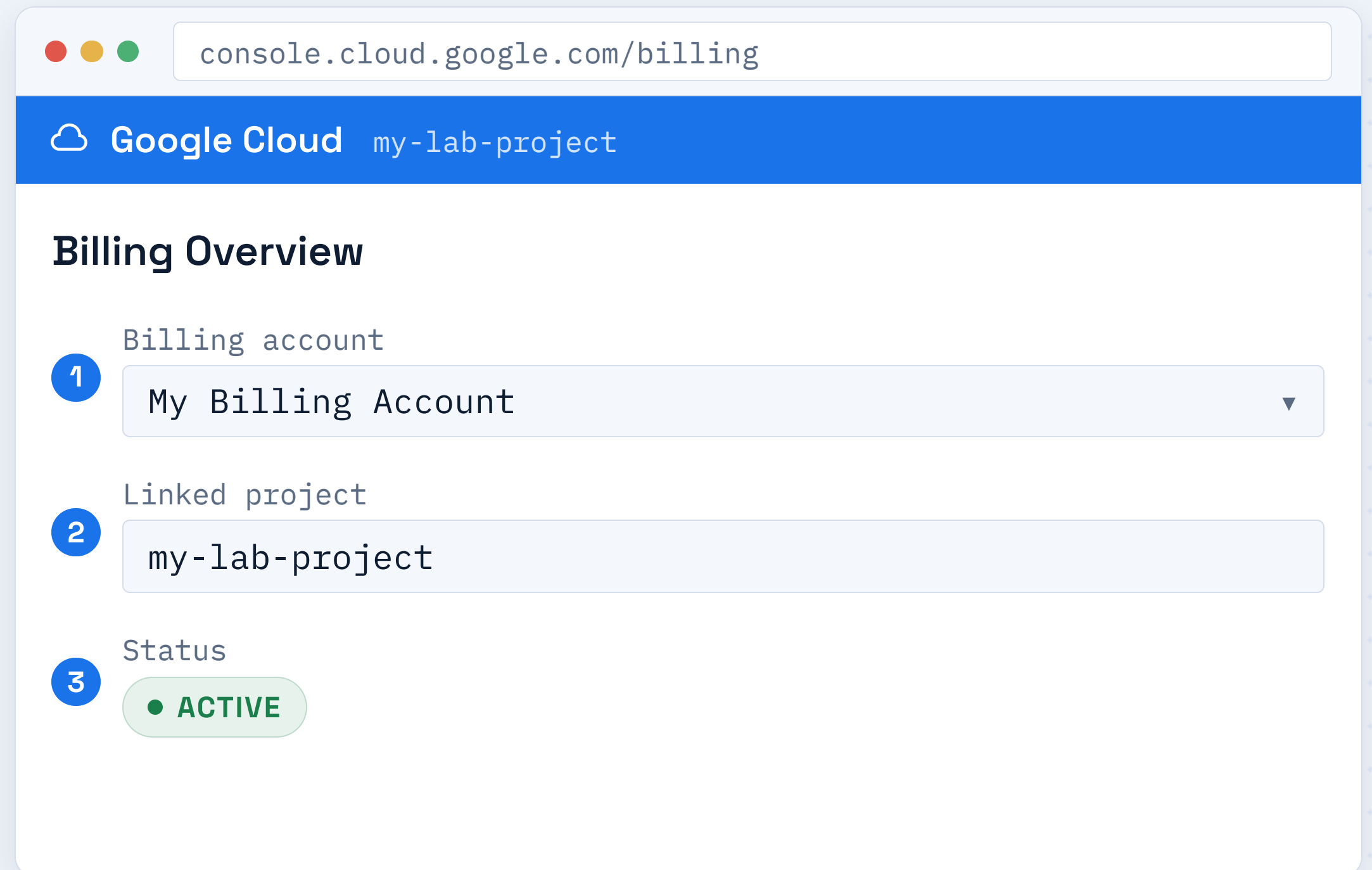
Resources

Confirm Billing Is Linked

STEPS

- 1 Open **Billing** from the navigation menu.
- 2 Check that a billing account is linked to your project.
- 3 Confirm the status shows **Active**.

✓ **Expected result:** billing shows Active and the project is linked.



Create & Connect to a Linux VM

Provision a virtual machine on Compute Engine, open a firewall path, and confirm you can reach it over SSH.



OBJECTIVE

What you will accomplish in this lab



STEPS

Numbered console actions to follow in order



EXPECTED RESULT

How to confirm the step worked



COMMON ERRORS

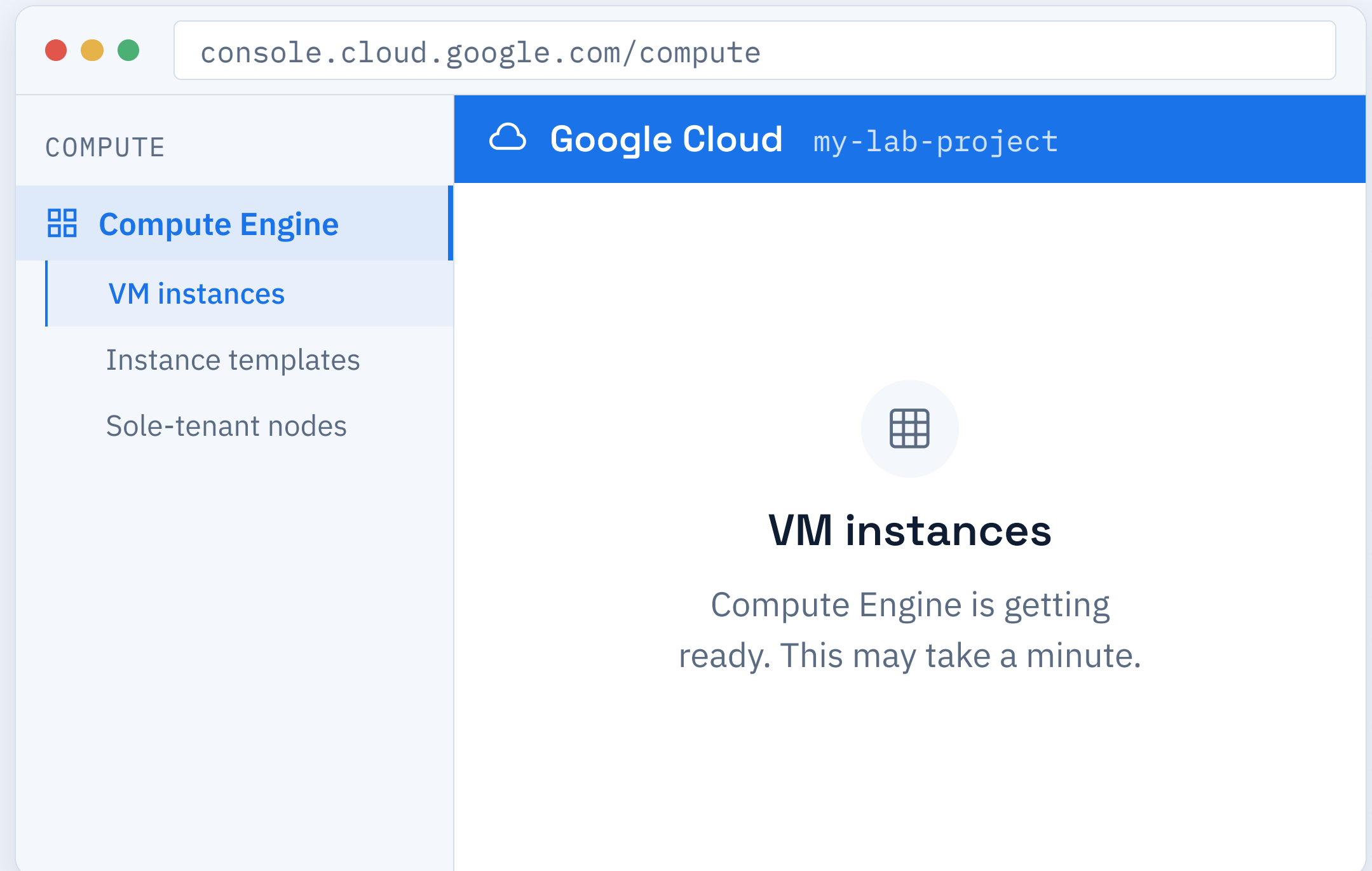
Pitfalls to watch for and how to fix them

Open Compute Engine

STEPS

- 1 Click the navigation menu (≡).
- 2 Under **Compute**, select **Compute Engine**.
- 3 Open the **VM instances** page.

✓ **Expected result:** the VM instances page loads (it may be empty at first).

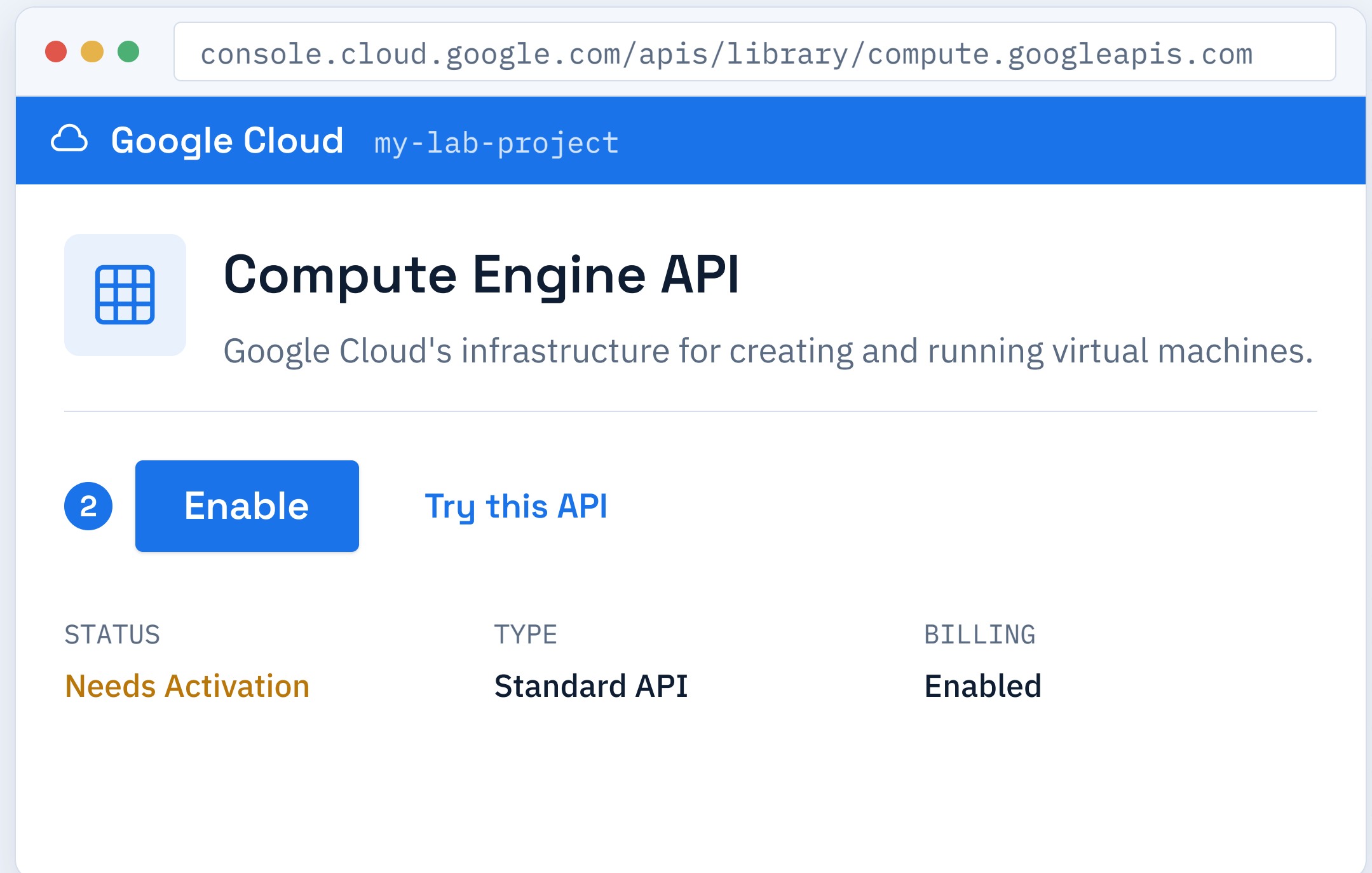


Enable the Compute Engine API

STEPS

- 1 The first visit prompts to enable the API.
- 2 Click **Enable**.
- 3 Wait for activation to finish.

✓ **Expected result:** the API shows Enabled and the instances page is ready.



console.cloud.google.com/apis/library/compute.googleapis.com

Google Cloud my-lab-project

Compute Engine API

Google Cloud's infrastructure for creating and running virtual machines.

2 **Enable** [Try this API](#)

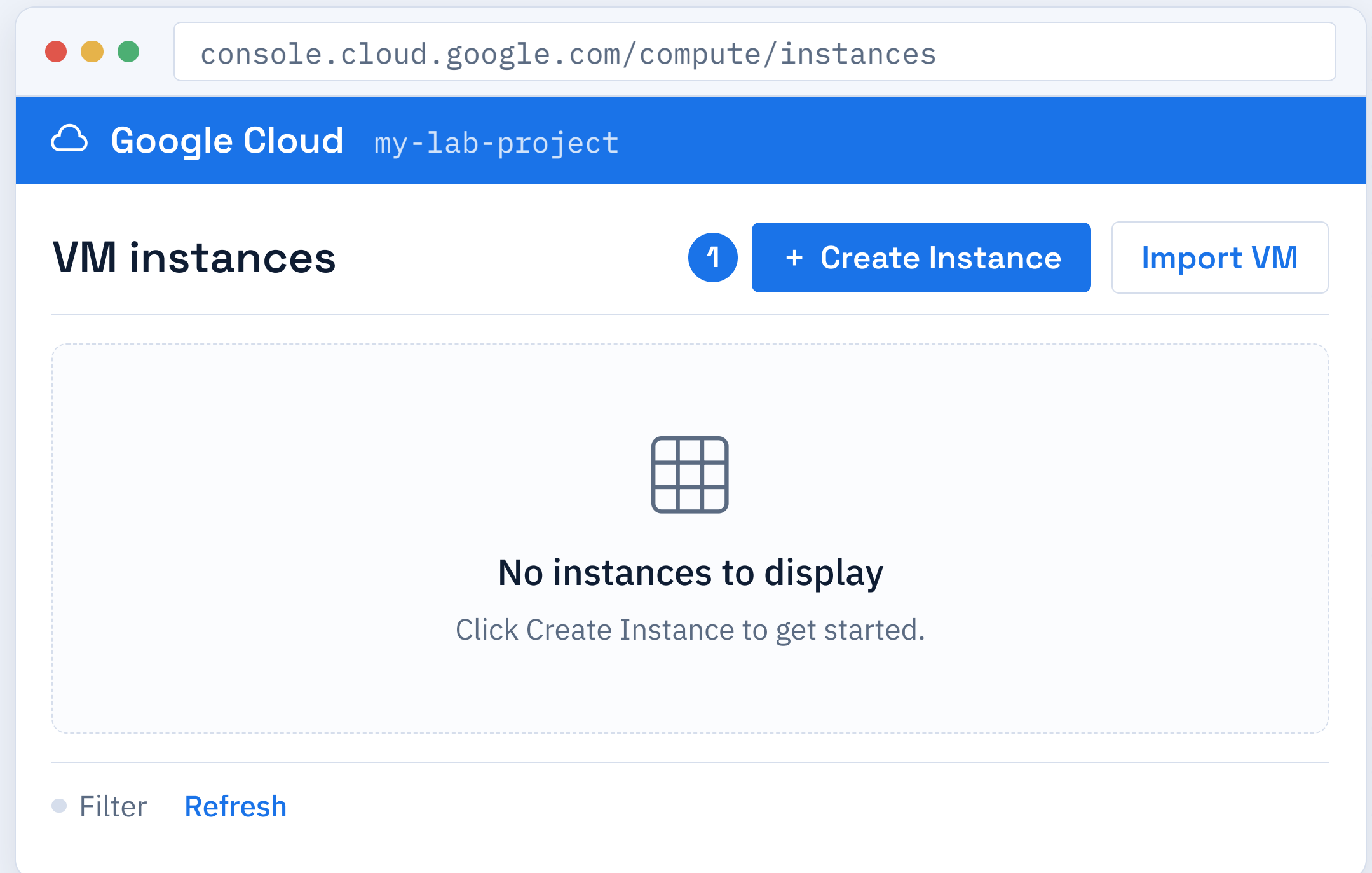
STATUS	TYPE	BILLING
Needs Activation	Standard API	Enabled

Start a New Instance

STEPS

- 1 On the VM instances page, click **Create Instance**.
- 2 The instance creation form opens.
- 3 You will fill it in over the next steps.

✓ **Expected result:** the Create an instance form is displayed.



Configure the VM Instance

STEPS

- 1 Enter a **Name** such as `lab-vm-01`.
- 2 Pick a **Region** close to you, e.g. `us-central1`.
- 3 Select a **Zone** within that region.

✓ **Expected result:** name and location fields are filled and the form is ready for machine type.



The screenshot shows a web browser window with the address bar displaying `console.cloud.google.com/compute/instancesAdd`. The browser's title bar shows the Google Cloud logo and the project name `my-lab-project`. The main content area is titled "Create an instance" and contains three numbered steps:

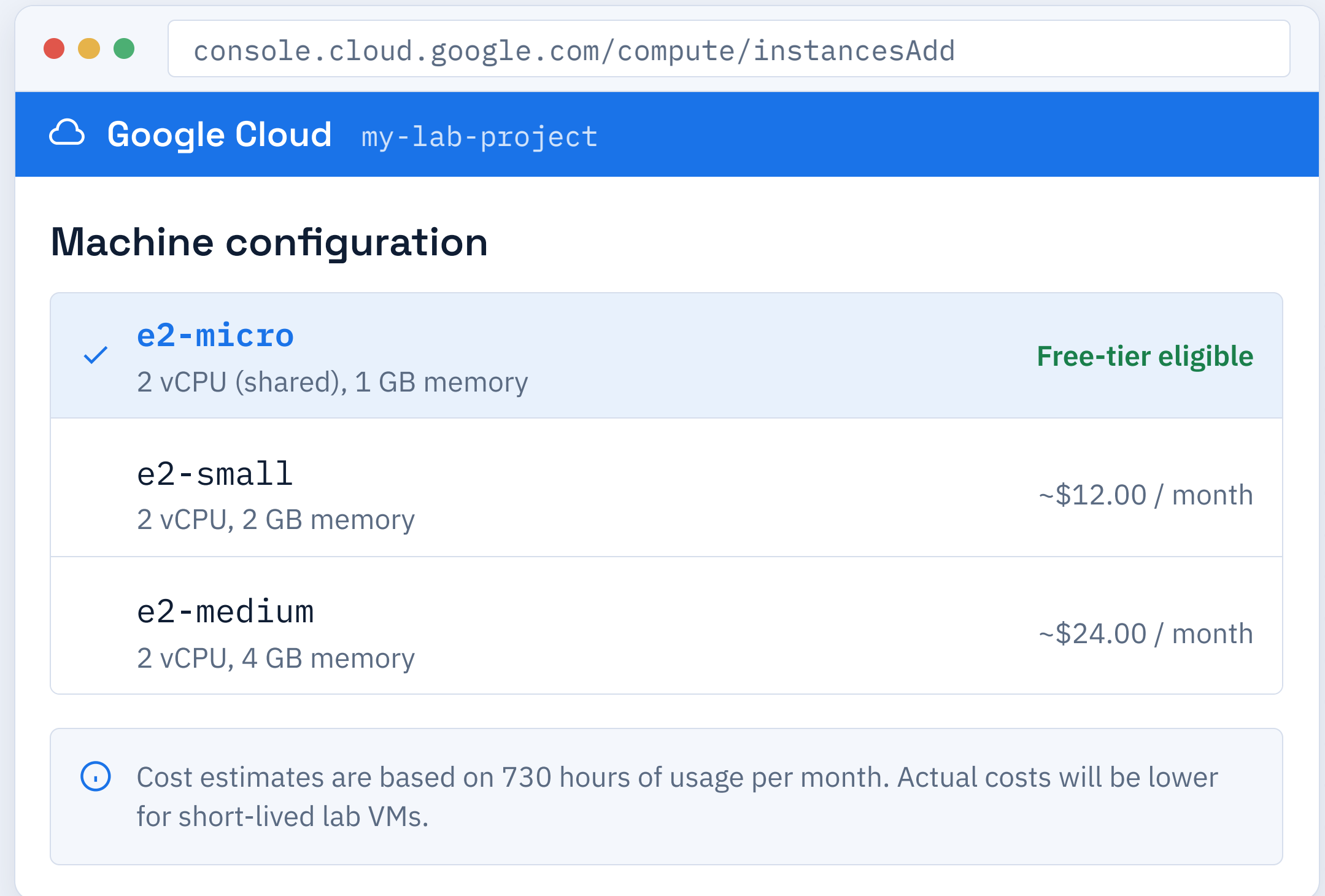
- 1 **Name**: A text input field containing `lab-vm-01`.
- 2 **Region**: A dropdown menu showing `us-central1 (Iowa)`.
- 3 **Zone**: A dropdown menu showing `us-central1-a`.

Choose a Machine Type

STEPS

- 1 Open the **Machine type** dropdown.
- 2 For labs, pick a small, low-cost type.
- 3 **e2-micro** is free-tier eligible.

✓ **Expected result:** a machine type is selected (e2-micro recommended).



The screenshot shows the Google Cloud console interface for adding a new instance. The browser address bar displays `console.cloud.google.com/compute/instancesAdd`. The header bar shows the Google Cloud logo and the project name `my-lab-project`. The main section is titled "Machine configuration" and lists three machine types: **e2-micro** (2 vCPU (shared), 1 GB memory) which is marked as "Free-tier eligible" with a green checkmark, **e2-small** (2 vCPU, 2 GB memory) with a cost estimate of ~\$12.00 / month, and **e2-medium** (2 vCPU, 4 GB memory) with a cost estimate of ~\$24.00 / month. A note at the bottom states: "Cost estimates are based on 730 hours of usage per month. Actual costs will be lower for short-lived lab VMs."

Machine Type	Configuration	Cost Estimate
e2-micro	2 vCPU (shared), 1 GB memory	Free-tier eligible
e2-small	2 vCPU, 2 GB memory	~\$12.00 / month
e2-medium	2 vCPU, 4 GB memory	~\$24.00 / month

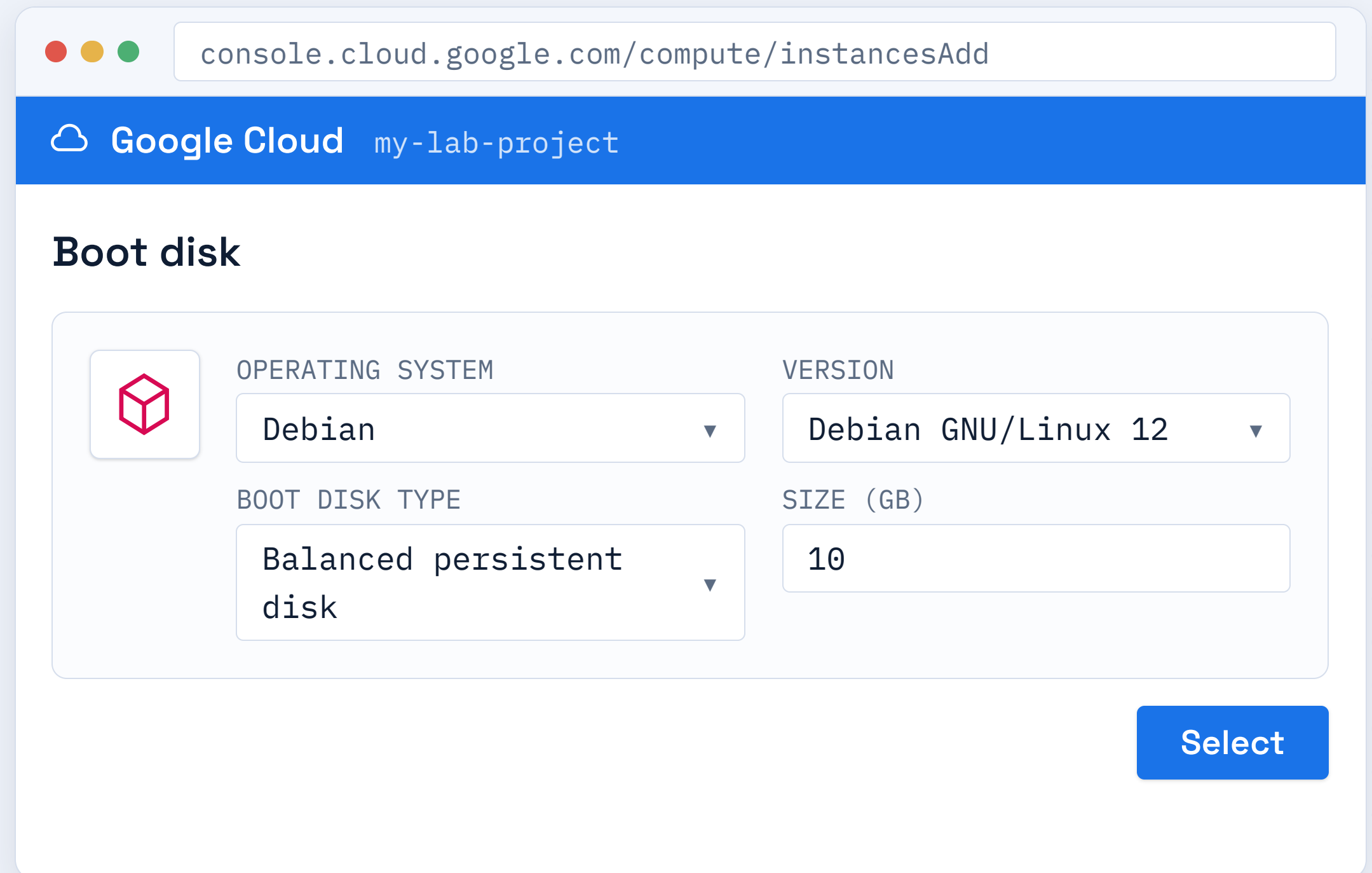
Cost estimates are based on 730 hours of usage per month. Actual costs will be lower for short-lived lab VMs.

Pick the Boot Disk Image

STEPS

- 1 Open the **Boot disk** section.
- 2 Choose an OS image, e.g. **Debian** or **Ubuntu**.
- 3 Keep the default disk size for labs.

✓ **Expected result:** the boot disk shows your chosen Linux image.



The screenshot shows the Google Cloud console interface for adding a new instance. The browser address bar displays `console.cloud.google.com/compute/instancesAdd`. The page header includes the Google Cloud logo and the project name `my-lab-project`. The 'Boot disk' section is active, showing a configuration card with a Debian icon. The configuration includes:

- OPERATING SYSTEM:** A dropdown menu set to 'Debian'.
- VERSION:** A dropdown menu set to 'Debian GNU/Linux 12'.
- BOOT DISK TYPE:** A dropdown menu set to 'Balanced persistent disk'.
- SIZE (GB):** A text input field containing the value '10'.

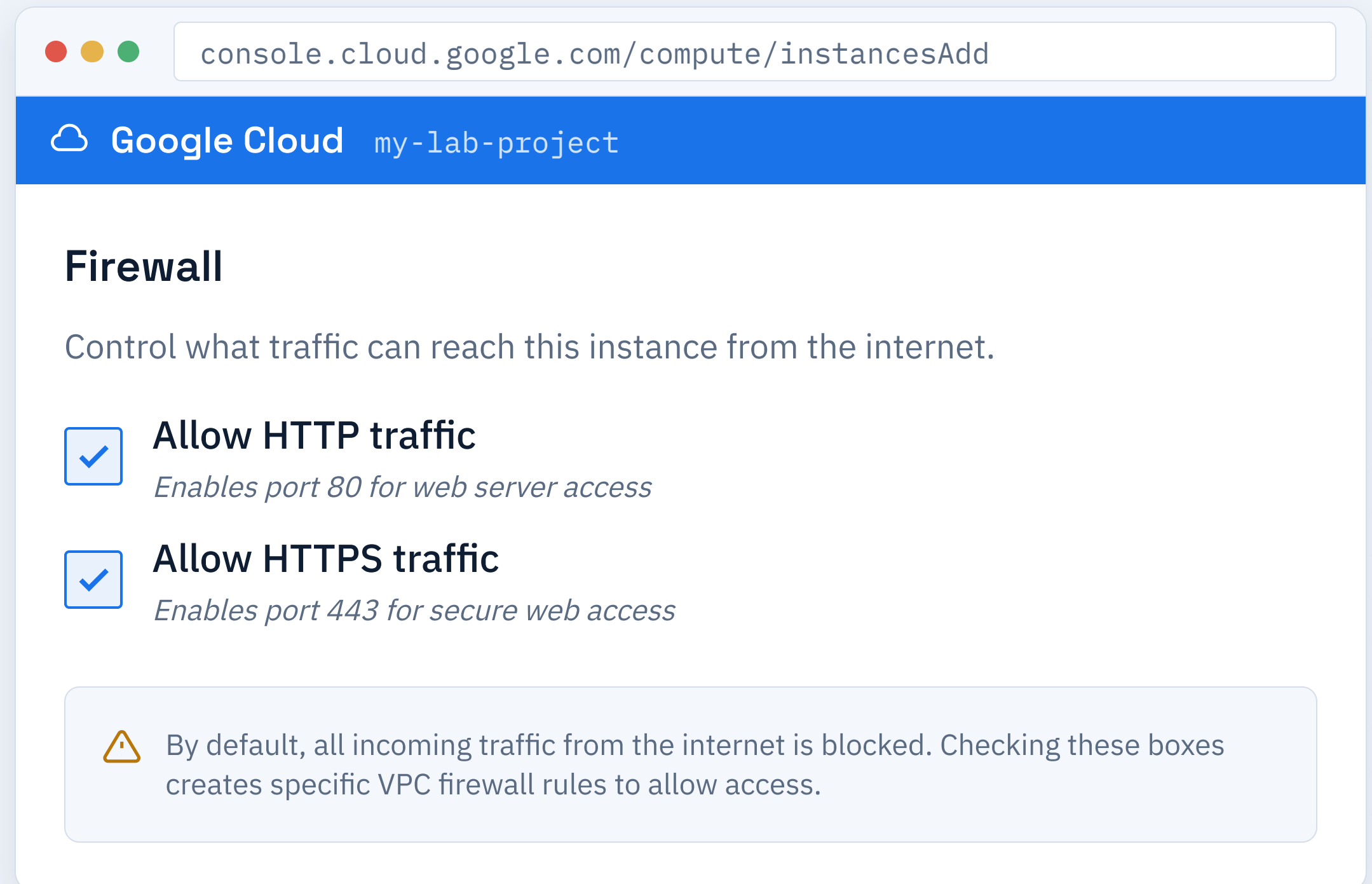
A blue 'Select' button is located at the bottom right of the configuration card.

Set Firewall Options

STEPS

- 1 Find the **Firewall** section in the form.
- 2 Check **Allow HTTP traffic**.
- 3 Check **Allow HTTPS traffic** if needed.

✓ **Expected result:** the HTTP (and HTTPS) firewall boxes are checked.



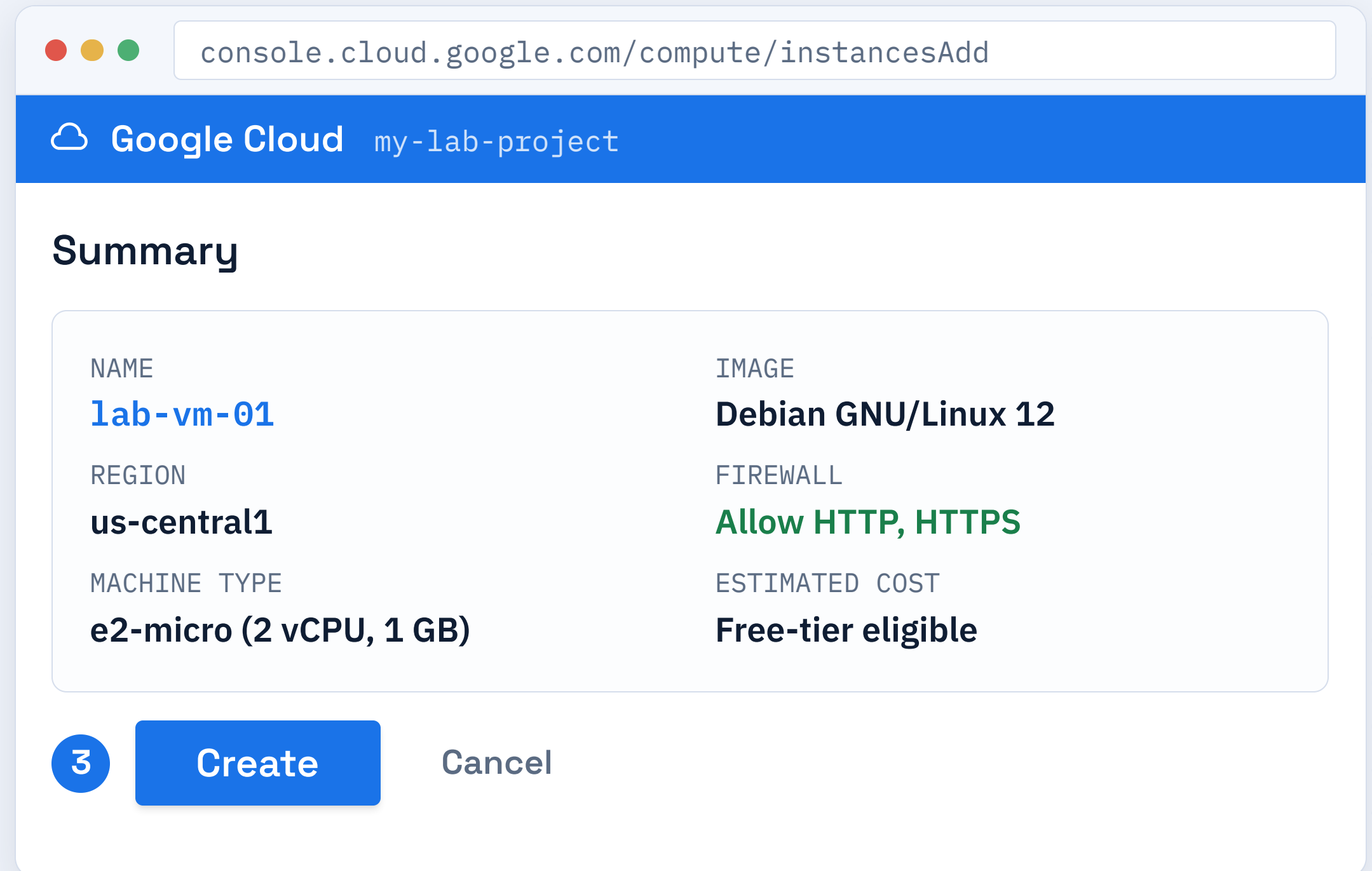
The screenshot shows the Google Cloud console interface for adding a new instance. The browser address bar displays `console.cloud.google.com/compute/instancesAdd`. The top navigation bar includes the Google Cloud logo and the project name `my-lab-project`. The main heading is **Firewall**, with a subtitle: "Control what traffic can reach this instance from the internet." Below this, there are two checked options: **Allow HTTP traffic** (with a subtext "Enables port 80 for web server access") and **Allow HTTPS traffic** (with a subtext "Enables port 443 for secure web access"). At the bottom, a warning box with a yellow triangle icon states: "By default, all incoming traffic from the internet is blocked. Checking these boxes creates specific VPC firewall rules to allow access."

Review and Create

STEPS

- 1 Review the instance summary.
- 2 Check name, region, machine type.
- 3 Click **Create**.

✓ **Expected result:** the VM begins provisioning and appears in the instances list.



The screenshot shows the Google Cloud console interface for creating a new VM instance. The browser address bar displays 'console.cloud.google.com/compute/instancesAdd'. The page header shows the Google Cloud logo and the project name 'my-lab-project'. The main section is titled 'Summary' and contains a table with the following details:

NAME	IMAGE
lab-vm-01	Debian GNU/Linux 12
REGION	FIREWALL
us-central1	Allow HTTP, HTTPS
MACHINE TYPE	ESTIMATED COST
e2-micro (2 vCPU, 1 GB)	Free-tier eligible

At the bottom of the summary section, there is a blue button labeled '3 Create' and a gray button labeled 'Cancel'.

Connect Over SSH

STEPS

- 1 In the instances list, find your VM.
- 2 Click the **SSH** button in its row.
- 3 A browser terminal opens a session.

✓ **Expected result:** a terminal opens and you see a shell prompt on the VM.

The screenshot shows the Google Cloud Platform console at the URL `console.cloud.google.com/compute/instances`. It displays a table of VM instances with the following data:

NAME	STATUS	CONNECT
lab-vm-01	● Running	2 SSH ▼

Below the table, a terminal window is open for the selected instance, showing the following output:

```
SSH: lab-vm-01  
  
Connecting to lab-vm-01...  
Linux lab-vm-01 6.1.0-13-cloud-amd64 #1 SMP Debian  
user@lab-vm-01 : ~ $
```

Verify the VM Is Running

STEPS

- 1 Return to the VM instances list.
- 2 Confirm the status is **Running** (green).
- 3 Note the **external IP** address.

✓ **Expected result:** the VM shows status Running with an external IP assigned.

console.cloud.google.com/compute/instances

VM instances

Filter: my-lab-project

NAME	ZONE	STATUS	EXTERNAL IP
lab-vm-01	us-central1-a	Running	34.31.25.114

VM Successfully Deployed

Your instance is live and ready for traffic on the external IP shown above.

When the VM Won't Cooperate

Quota exceeded

Free-tier projects limit CPUs per region.

Fix: choose a smaller machine type or a different region.

API not enabled

The Compute Engine API must be on first.

Fix: open APIs & Services and click Enable.

SSH connection refused

Firewall blocks port 22 or the VM is still booting.

Fix: allow port 22 and wait for status Running.

Billing not linked

Resources cannot be created without billing.

Fix: attach a billing account to the project.

How Cloud Networking Works

A few building blocks control how your VM is reached and how it communicates with the world.



VPC NETWORK

A private virtual network for your resources.



SUBNET

A regional IP range inside the VPC.



IP ADDRESSES

Internal for private traffic, external for the internet.



FIREWALL RULES

Allow or deny traffic by port and protocol.

External IP Addresses

An external IP makes a VM reachable from the internet. There are two main types to consider.



EPHEMERAL

Assigned automatically, changes when the instance is stopped and restarted.



STATIC

Reserved and fixed. The IP stays the same even if the VM is deleted or moved.

Internal IP Addresses

Internal IPs let resources talk privately inside the VPC without exposing traffic to the internet.



PRIVATE RANGE

From the subnet's internal IP range (e.g., 10.128.0.0/20), not internet-routable.



VM-TO-VM

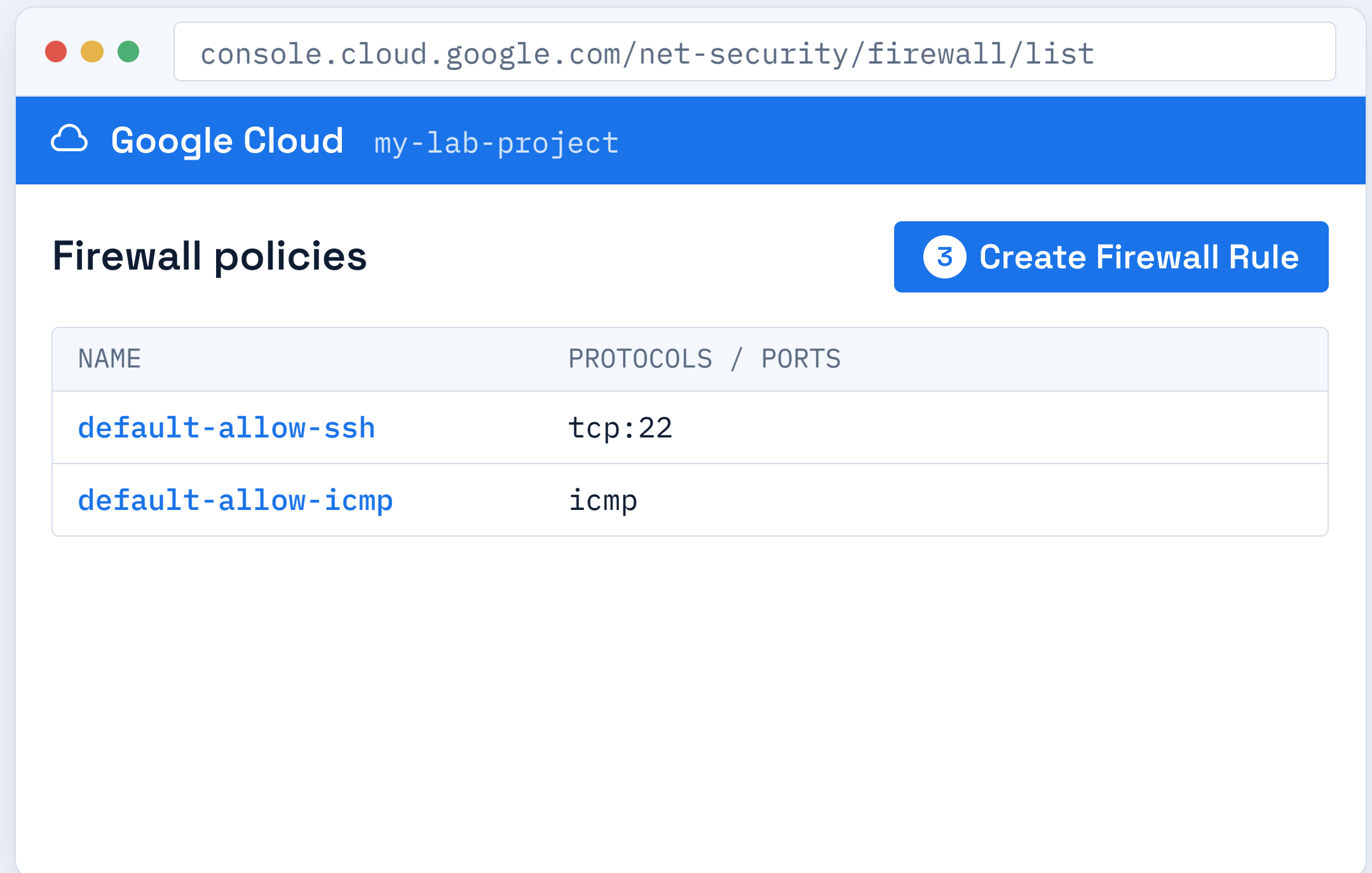
Used for low-latency traffic between instances in the same VPC or peered networks.

Open the Firewall Rules Page

STEPS

- 1 In the navigation menu, go to **VPC network** → **Firewall**.
- 2 Review the **default rules** already present in the list.
- 3 Click the **Create Firewall Rule** button at the top.

✓ **Expected result:** the firewall rules list is open and ready for a new rule.



Allow HTTP Traffic

STEPS

- 1 Name the rule **allow-http**.
- 2 Set **Targets** to "All instances in the network" and **Source ranges** to **0.0.0.0/0**.
- 3 In **Protocols and ports**, check **TCP** and enter **80**.

✓ **Expected result:** a rule allowing incoming traffic on port 80 is created.

console.cloud.google.com/net-security/firewall/add

Google Cloud

my-lab-project

Create a firewall rule

1

Name

allow-http

2

Targets

All instances in the network ▼

Source IPv4 ranges

0.0.0.0/0

3

Protocols and ports

☒ TCP

80

Create

Allow HTTPS Traffic

STEPS

- 1 Create another rule named **allow-https**.
- 2 Set **Source ranges** to **0.0.0.0/0** for universal access.
- 3 In **Protocols and ports**, enter **443** for TCP.

✓ **Expected result:** a rule allowing incoming secure traffic on port 443 is created.

console.cloud.google.com/net-security/firewall/add

Google Cloud

my-lab-project

Create a firewall rule

1

Name

allow-https

2

Source IPv4 ranges

0.0.0.0/0

3

Protocols and ports

☒ TCP

443

Create

Networking Gotchas

Forgot the firewall rule

Traffic is blocked by default even if the service is running.

Fix: create an allow rule for the specific port you need.

Wrong source range

Rule scoped too narrowly (e.g. only your home IP).

Fix: use 0.0.0.0/0 for public access during labs.

No external IP

VM created without one cannot be reached from the internet.

Fix: assign an external IP (ephemeral or static) to the instance.

Port mismatch

Service runs on a different port than the firewall allows.

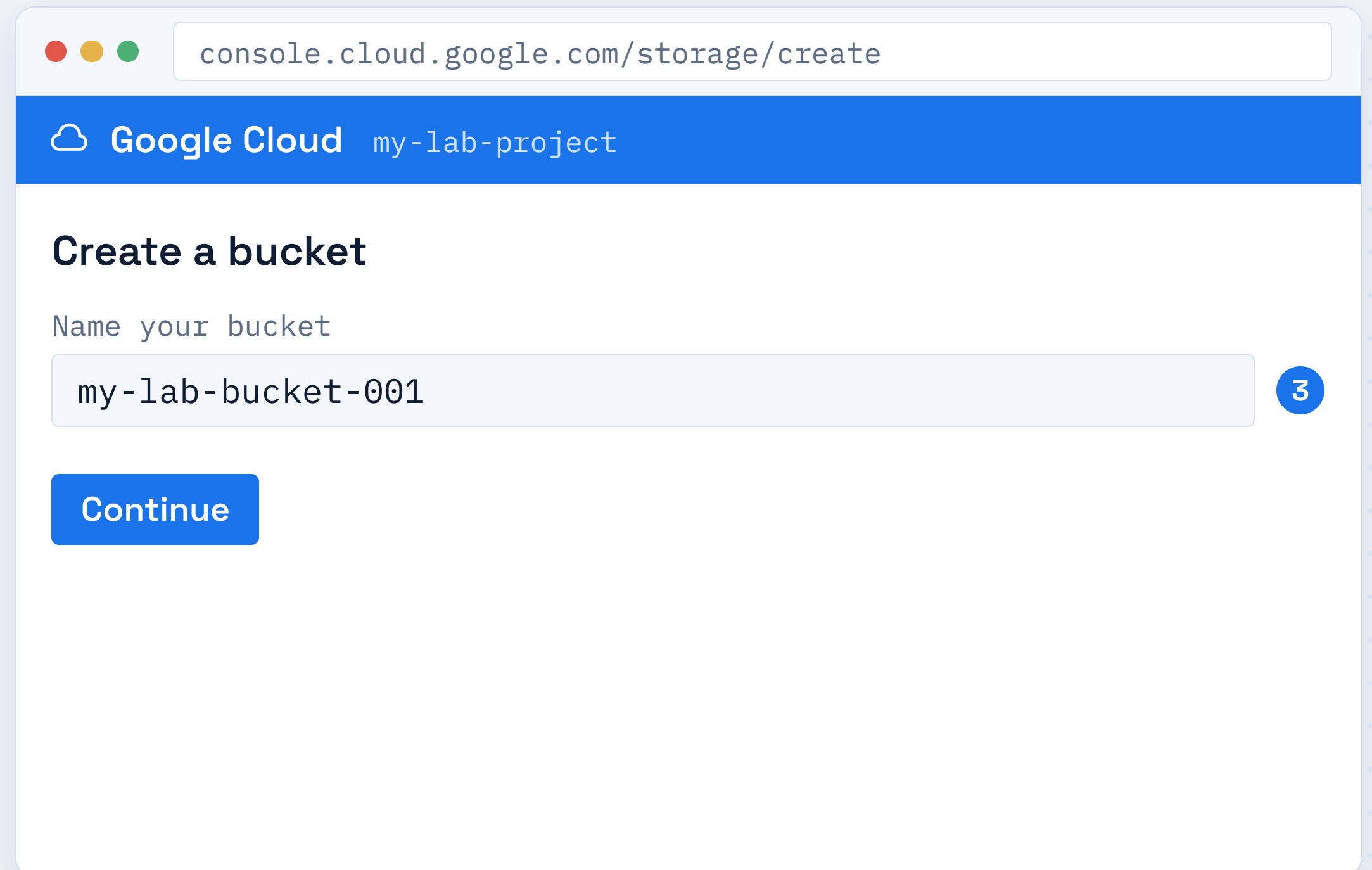
Fix: open the exact port your application is listening on.

Create a Storage Bucket

STEPS

- 1 Open **Cloud Storage** → **Buckets**.
- 2 Click **Create**.
- 3 Enter a globally unique bucket name.

✓ **Expected result:** the bucket creation form opens with your name accepted.



The screenshot shows a web browser window with the address bar containing 'console.cloud.google.com/storage/create'. The page header is blue with the Google Cloud logo and the text 'my-lab-project'. The main heading is 'Create a bucket'. Below it, the text 'Name your bucket' is followed by a text input field containing 'my-lab-bucket-001'. A blue circle with the number '3' is positioned to the right of the input field. Below the input field is a blue button labeled 'Continue'.

Bucket Naming Rules

Buckets follow strict naming conventions enforced across the entire global service.



GLOBALLY UNIQUE

No two buckets share a name across all of GCP.



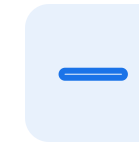
LOWERCASE

Letters, numbers, dashes; no uppercase.



LENGTH

3 to 63 characters in total length.



NO SPACES

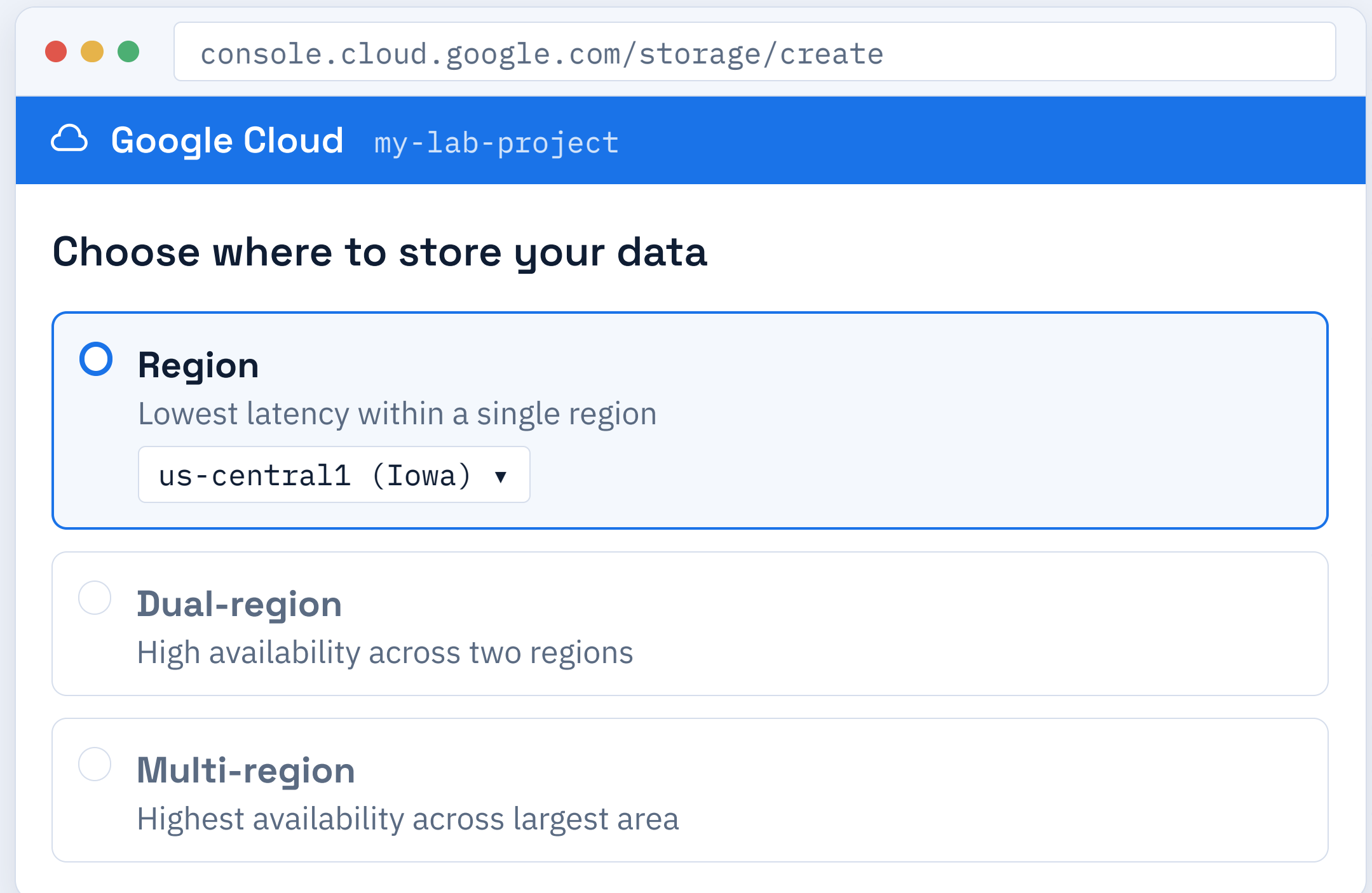
Use dashes, not spaces or special characters.

Choose Where Data Lives

STEPS

- 1 Pick a **Location type**.
- 2 **Region** keeps data in one place for lowest cost.
- 3 **Multi-region** spreads data for high availability.

✓ **Expected result:** a location is selected (Region recommended for labs).



console.cloud.google.com/storage/create

Google Cloud my-lab-project

Choose where to store your data

- ☒ **Region**
Lowest latency within a single region
us-central1 (Iowa) ▼
- ☐ **Dual-region**
High availability across two regions
- ☐ **Multi-region**
Highest availability across largest area

Pick a Storage Class

Storage classes balance data availability and performance with cost.



STANDARD

Frequent access, hot data, highest performance.



NEARLINE

Access about once a month, lower storage cost.



COLDLINE

Access about once a quarter, deep storage discount.



ARCHIVE

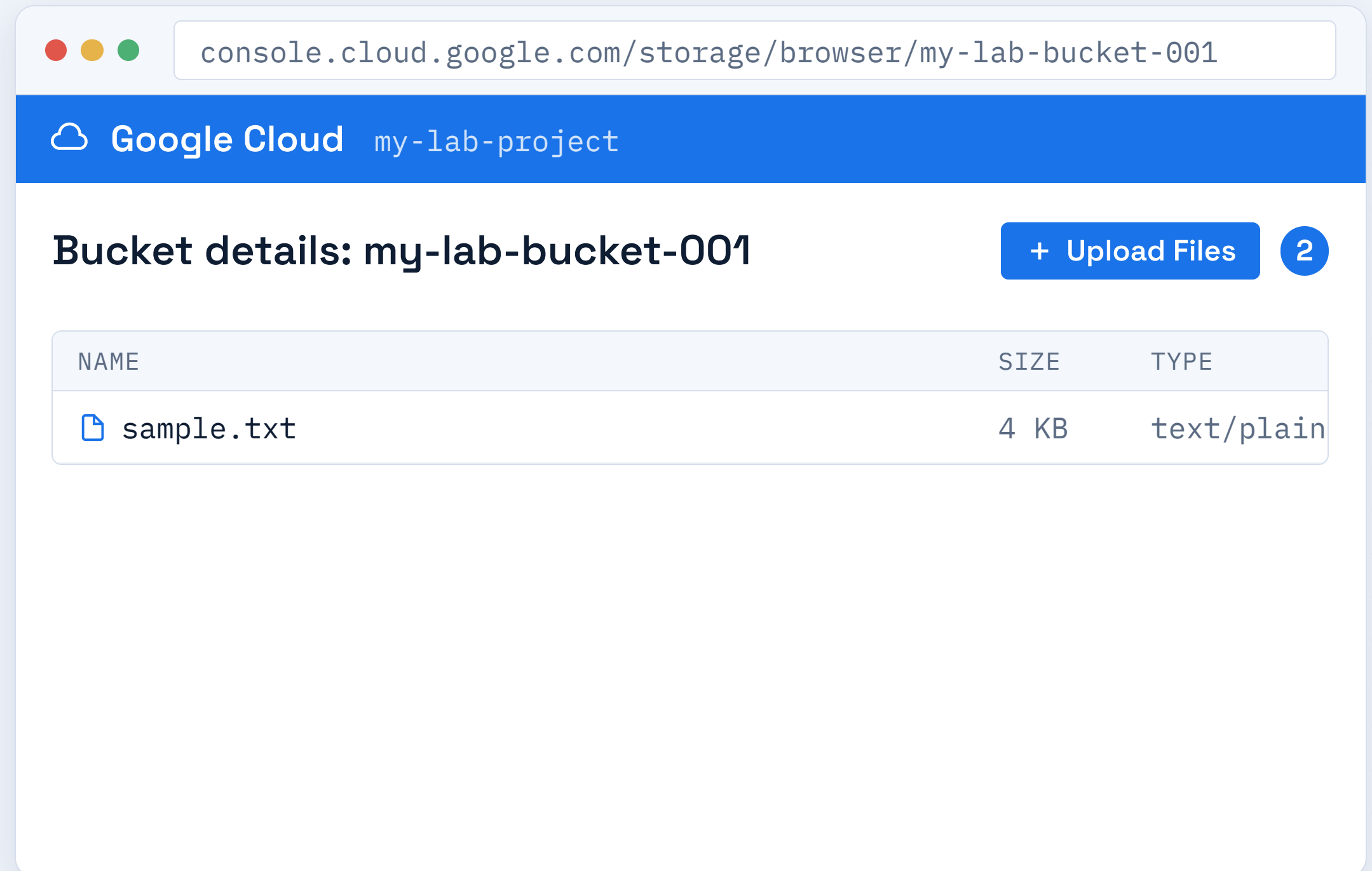
Rare access (once a year), lowest storage cost.

Upload an Object

STEPS

- 1 Open your bucket from the list.
- 2 Click **Upload Files**.
- 3 Select a file from your local computer.

✓ **Expected result:** the file appears in the bucket's object list.

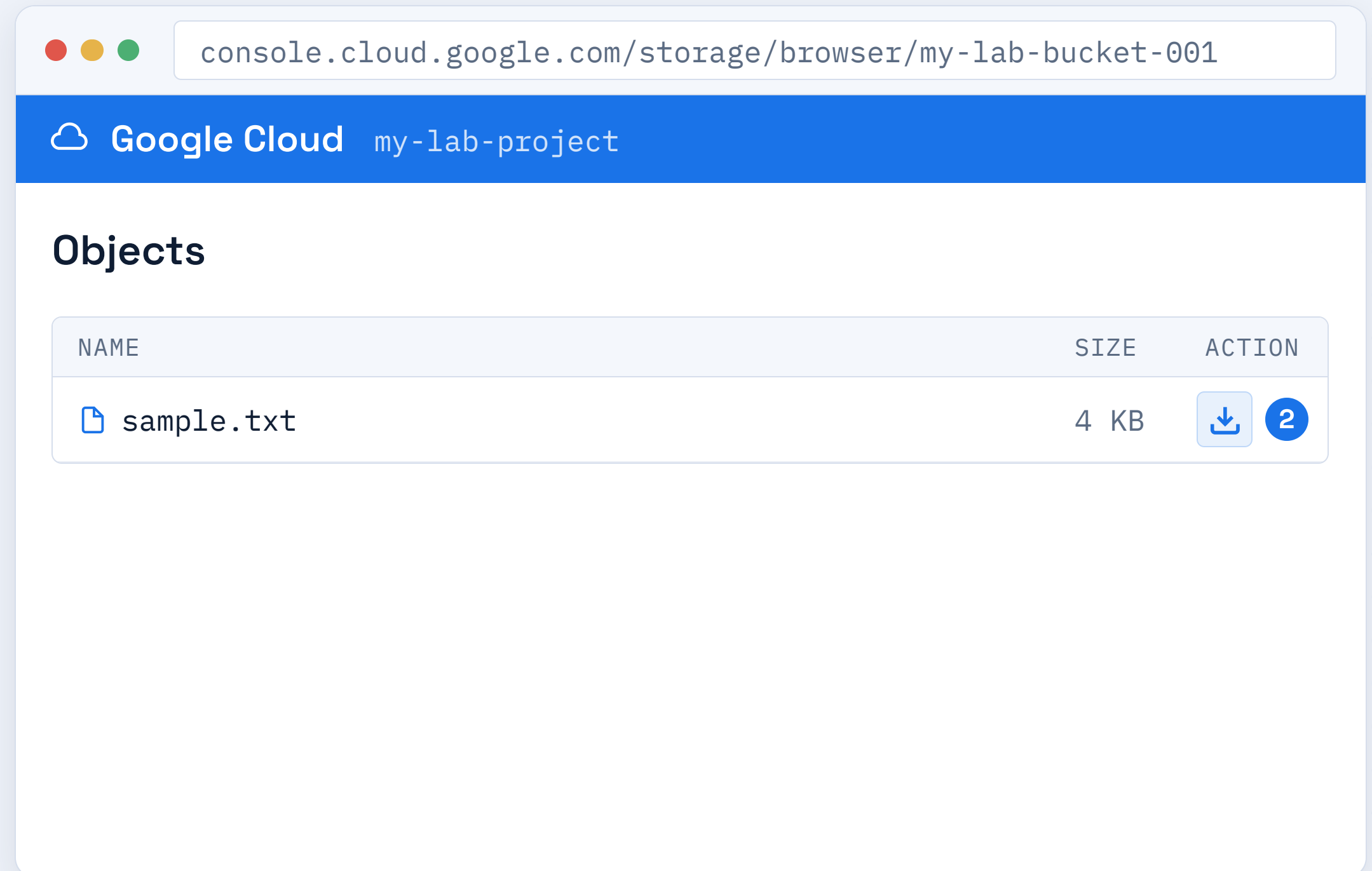


Download an Object

STEPS

- 1 Find the object in the bucket's list.
- 2 Click the **download icon** in its row.
- 3 The file saves locally to your device.

✓ **Expected result:** the object downloads to your computer.

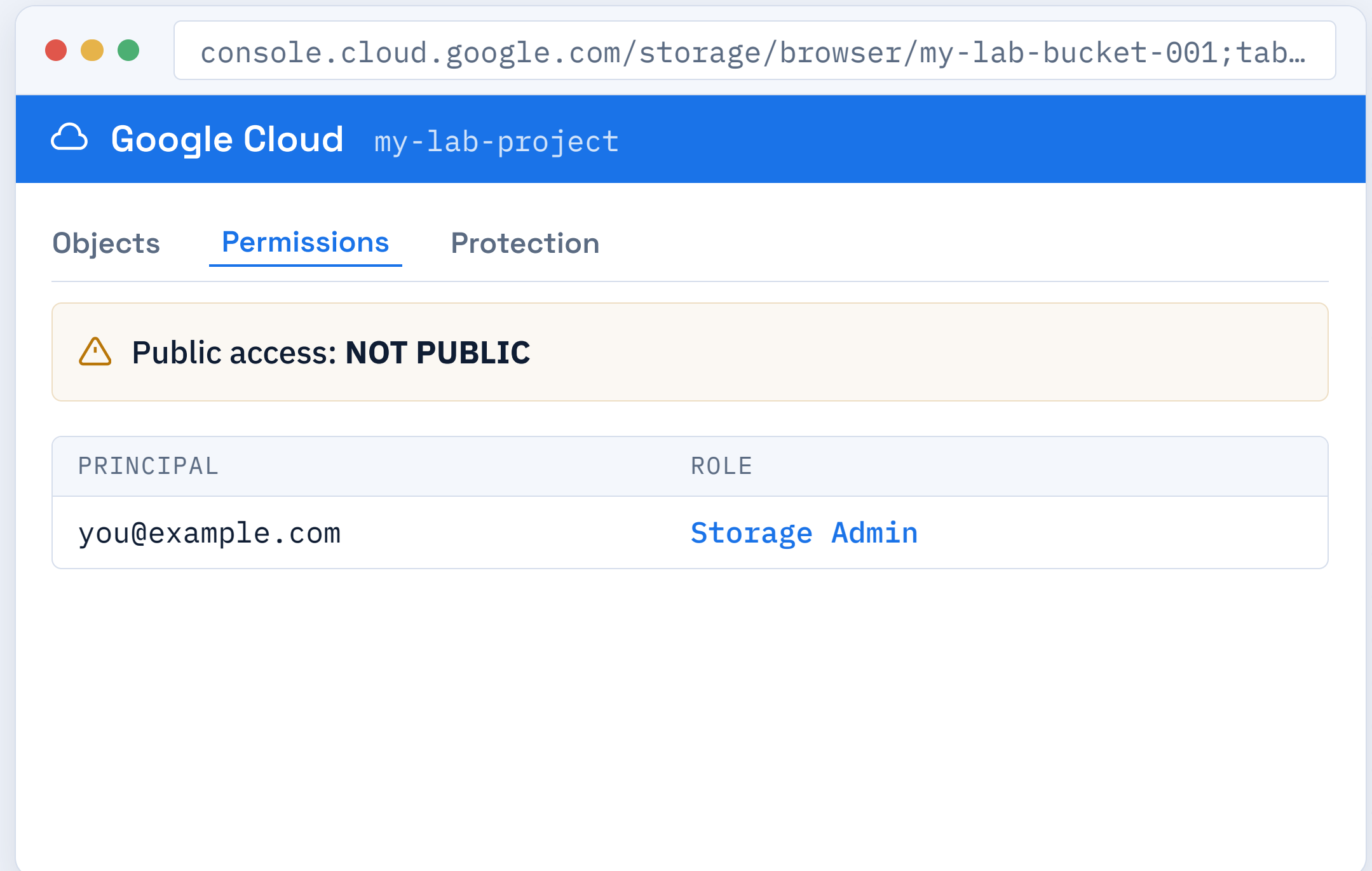


Control Who Can Access

STEPS

- 1 Open the **Permissions** tab.
- 2 Review IAM roles assigned to the bucket.
- 3 Avoid making data public unless required.

✓ **Expected result:** you can see the principals and roles on the bucket.



Storage Pitfalls

Bucket name taken

Bucket names must be globally unique across all of GCP.

Fix: add a unique suffix such as your name or a random number.

Invalid name

Names can only contain lowercase letters, numbers, and dashes.

Fix: use lowercase letters and dashes only; avoid spaces.

Access denied

The principal is missing the required IAM permission for the bucket.

Fix: grant the correct Storage Object Viewer or Admin role.

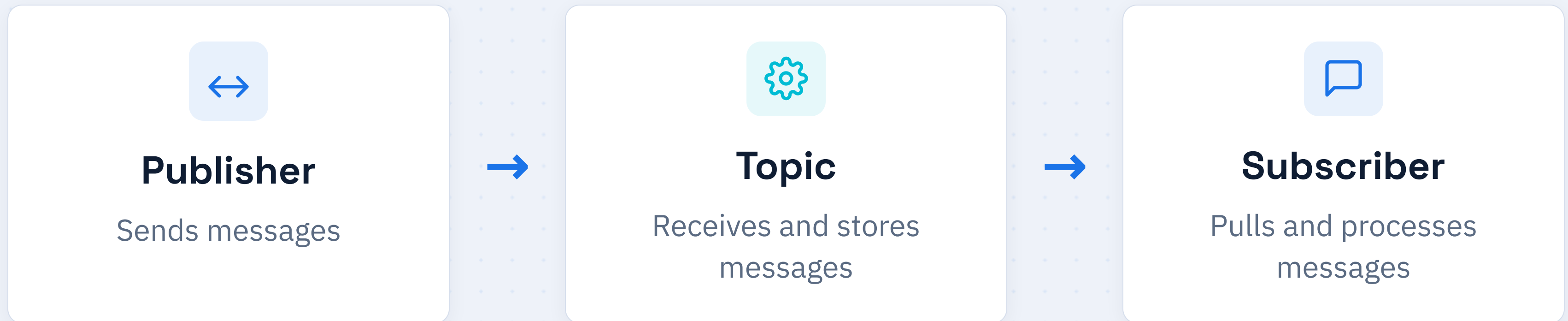
Accidentally public

The bucket has been granted read access to "allUsers".

Fix: remove the public access entry from the permissions list.

The Messaging Pipeline

Pub/Sub decouples senders from receivers.

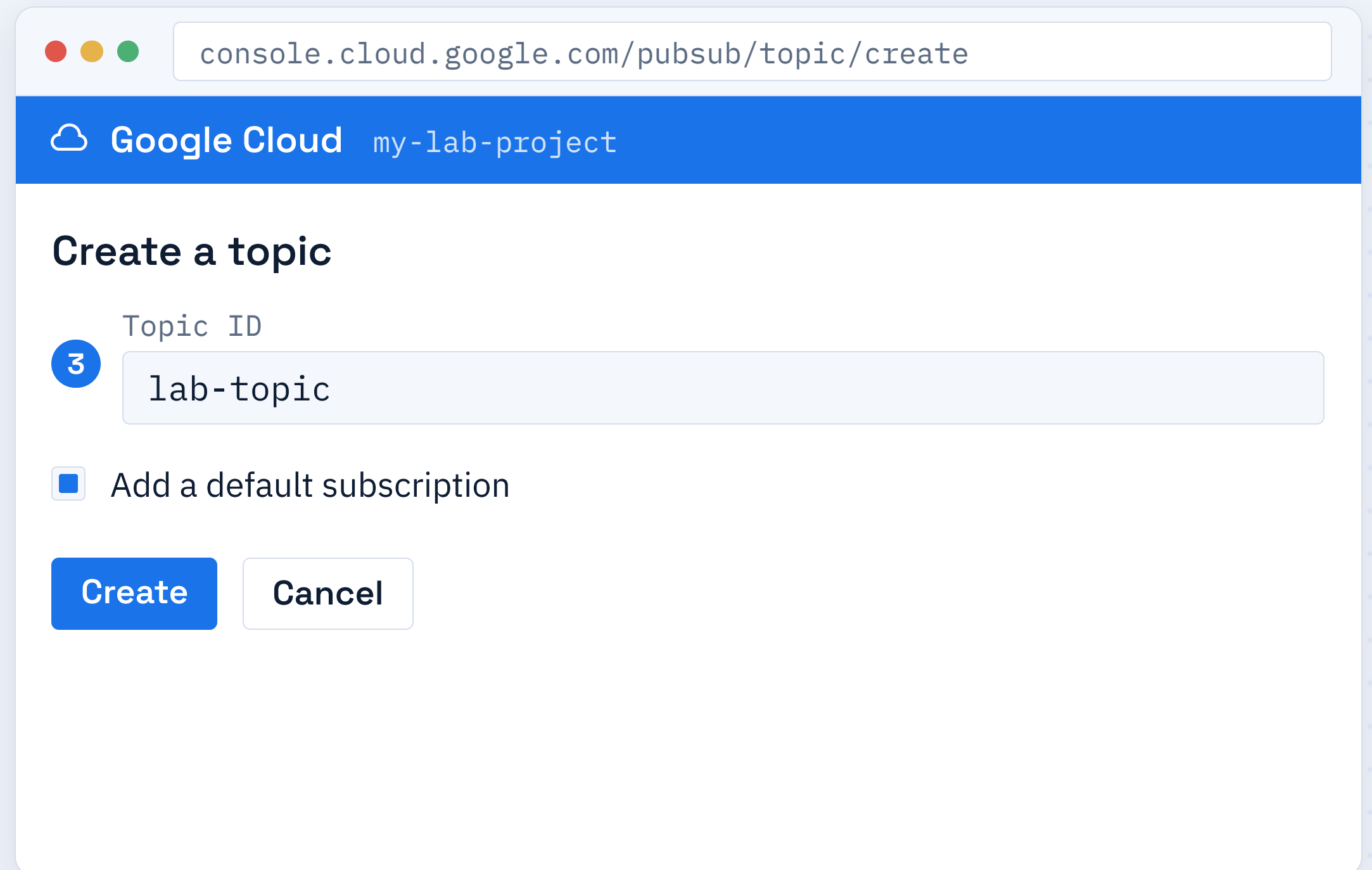


Create a Topic

STEPS

- 1 Open **Pub/Sub** → **Topics**.
- 2 Click **Create Topic**.
- 3 Enter a **Topic ID**, e.g. lab-topic.

✓ **Expected result:** the topic appears in the topics list.



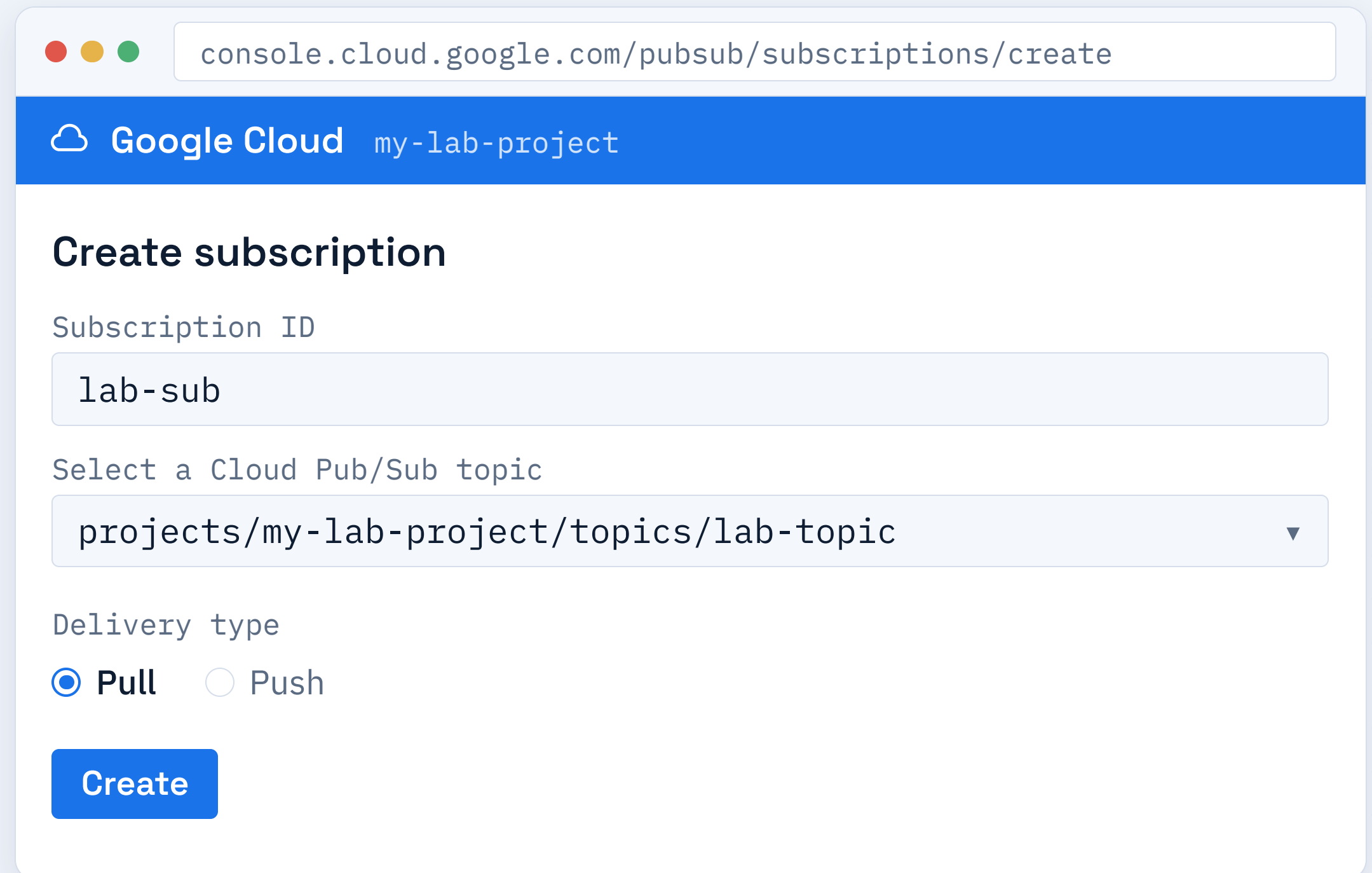
The screenshot shows a web browser window with the address bar displaying 'console.cloud.google.com/pubsub/topic/create'. The browser's title bar shows 'Google Cloud my-lab-project'. The main content area is titled 'Create a topic'. Below the title, there is a label 'Topic ID' and a text input field containing 'lab-topic'. A blue circle with the number '3' is positioned to the left of the input field. Below the input field, there is a checkbox labeled 'Add a default subscription', which is currently unchecked. At the bottom of the form, there are two buttons: a blue 'Create' button and a white 'Cancel' button with a grey border.

Create a Subscription

STEPS

- 1 Open your topic and scroll to the **Subscriptions** section.
- 2 Click **Create Subscription**.
- 3 Choose **Pull** delivery type and click Create.

✓ **Expected result:** a pull subscription is attached to the topic.



The screenshot shows the 'Create subscription' page in the Google Cloud console. The browser address bar displays 'console.cloud.google.com/pubsub/subscriptions/create'. The page header includes the Google Cloud logo and the project name 'my-lab-project'. The main heading is 'Create subscription'. Below this, there are three form fields: 'Subscription ID' with the value 'lab-sub', 'Select a Cloud Pub/Sub topic' with the value 'projects/my-lab-project/topics/lab-topic', and 'Delivery type' with 'Pull' selected (indicated by a blue radio button) and 'Push' as an option. A blue 'Create' button is located at the bottom of the form.

console.cloud.google.com/pubsub/subscriptions/create

Google Cloud my-lab-project

Create subscription

Subscription ID

lab-sub

Select a Cloud Pub/Sub topic

projects/my-lab-project/topics/lab-topic ▼

Delivery type

☒ Pull ☐ Push

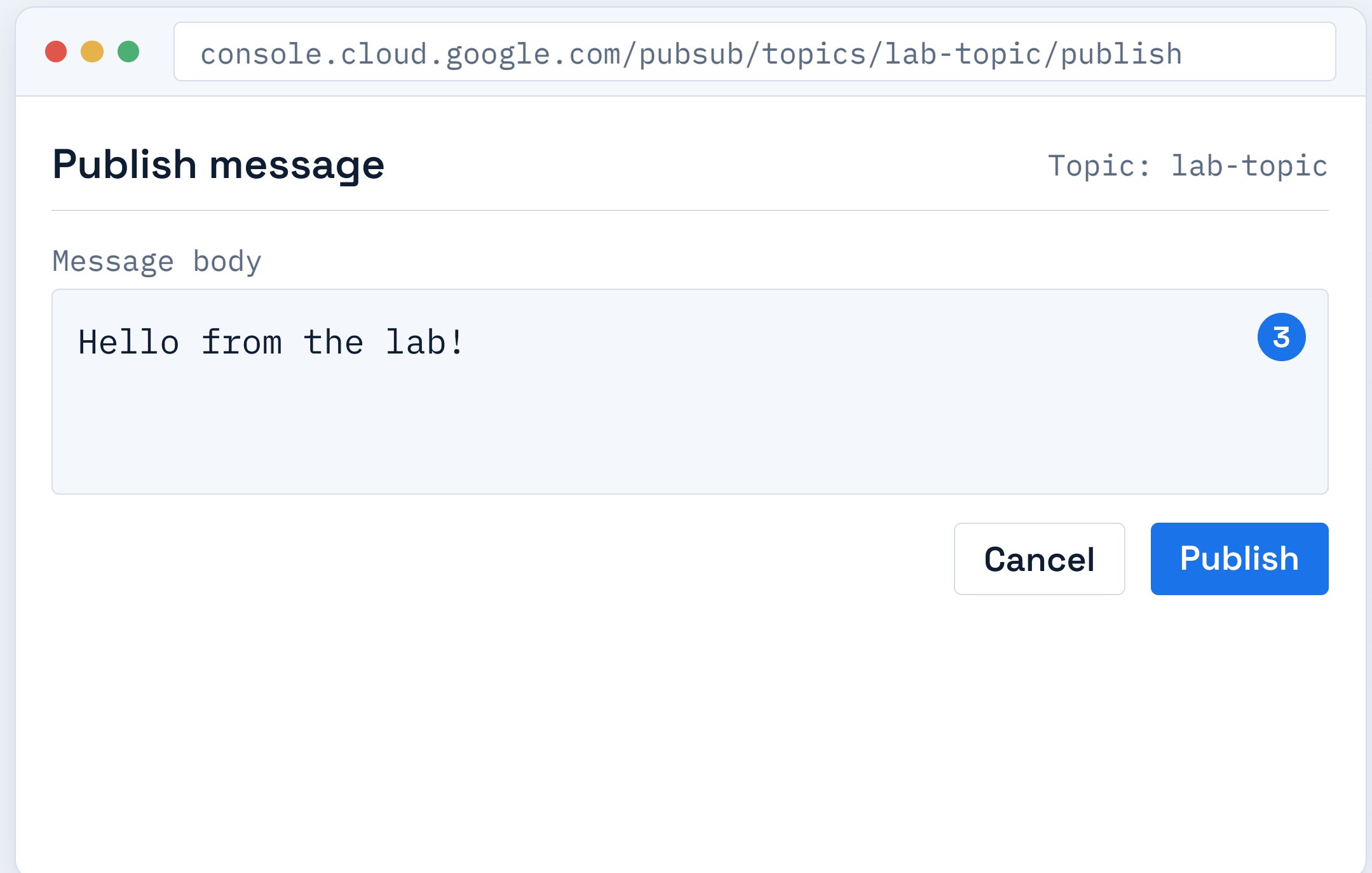
Create

Publish a Message

STEPS

- 1 Open the **lab-topic** and click the **Messages** tab.
- 2 Click **Publish Message**.
- 3 Type a message body and click **Publish**.

✓ **Expected result:** the message is accepted by the topic.



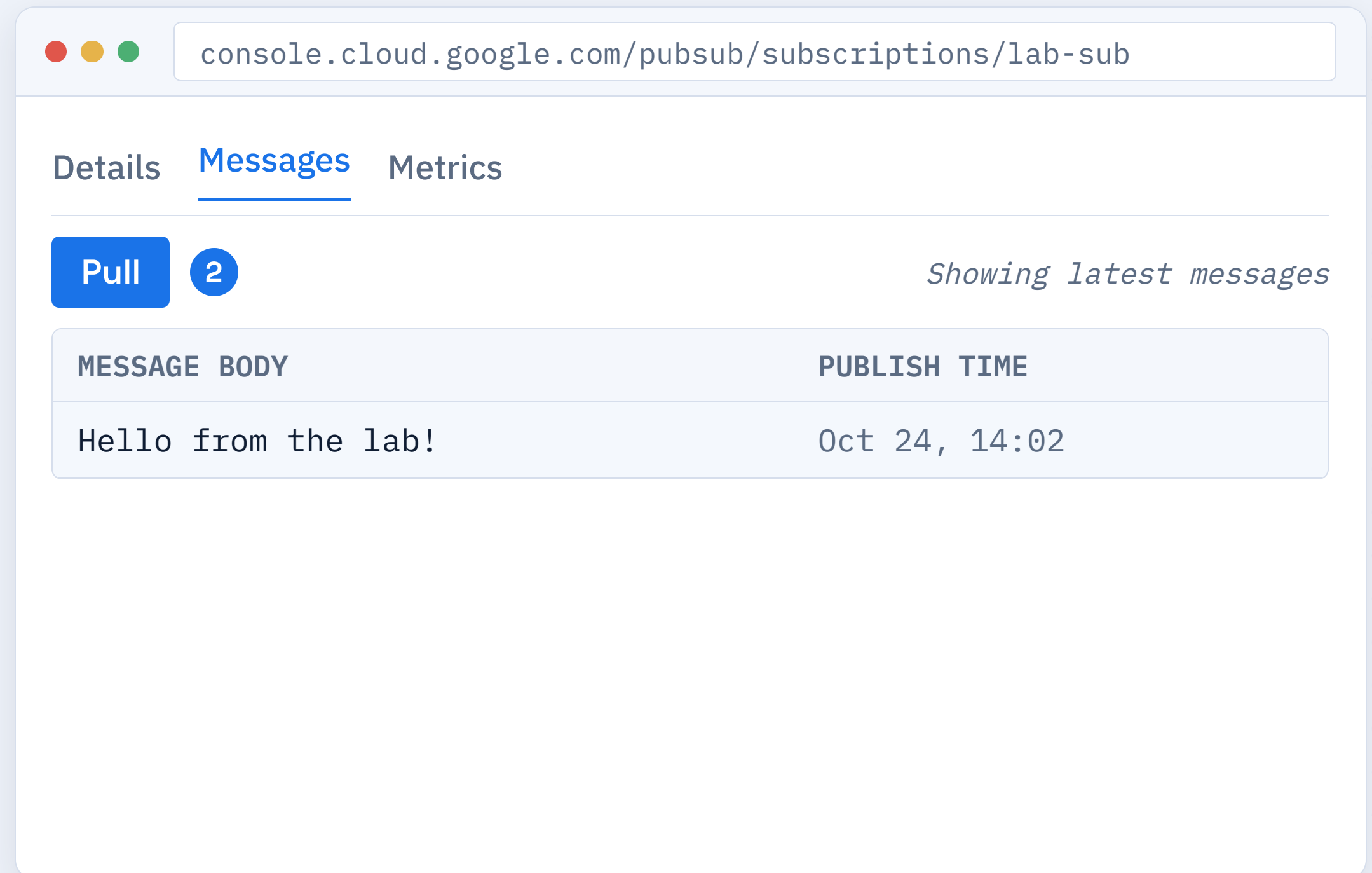
The screenshot shows a web browser window with the address bar containing the URL `console.cloud.google.com/pubsub/topics/lab-topic/publish`. The main content area is titled "Publish message" and includes a label "Topic: lab-topic" on the right. Below the title is a section labeled "Message body" containing a text input field with the text "Hello from the lab!". A blue circle with the number "3" is positioned at the end of the text input field. At the bottom right of the dialog are two buttons: "Cancel" and "Publish".

Pull the Message

STEPS

- 1 Open the **lab-sub** subscription.
- 2 Click the **Pull** button in the Messages tab.
- 3 The message body appears in the results list.

✓ **Expected result:** the published message is shown in the pull results.



console.cloud.google.com/pubsub/subscriptions/lab-sub

Details Messages Metrics

Pull 2 *Showing latest messages*

MESSAGE BODY	PUBLISH TIME
Hello from the lab!	Oct 24, 14:02

Verify Delivery

STEPS

- 1 Confirm the pulled body matches what you published.
- 2 **Acknowledge** the message to remove it from the queue.
- 3 Check that the subscription has no backlog remaining.

✓ **Expected result:** the message body matches and the backlog is zero.

console.cloud.google.com/pubsub/subscriptions/lab-sub

Pull results

● Unacked: 0

MESSAGE BODY	ACTION
Hello from the lab!	Ack 2

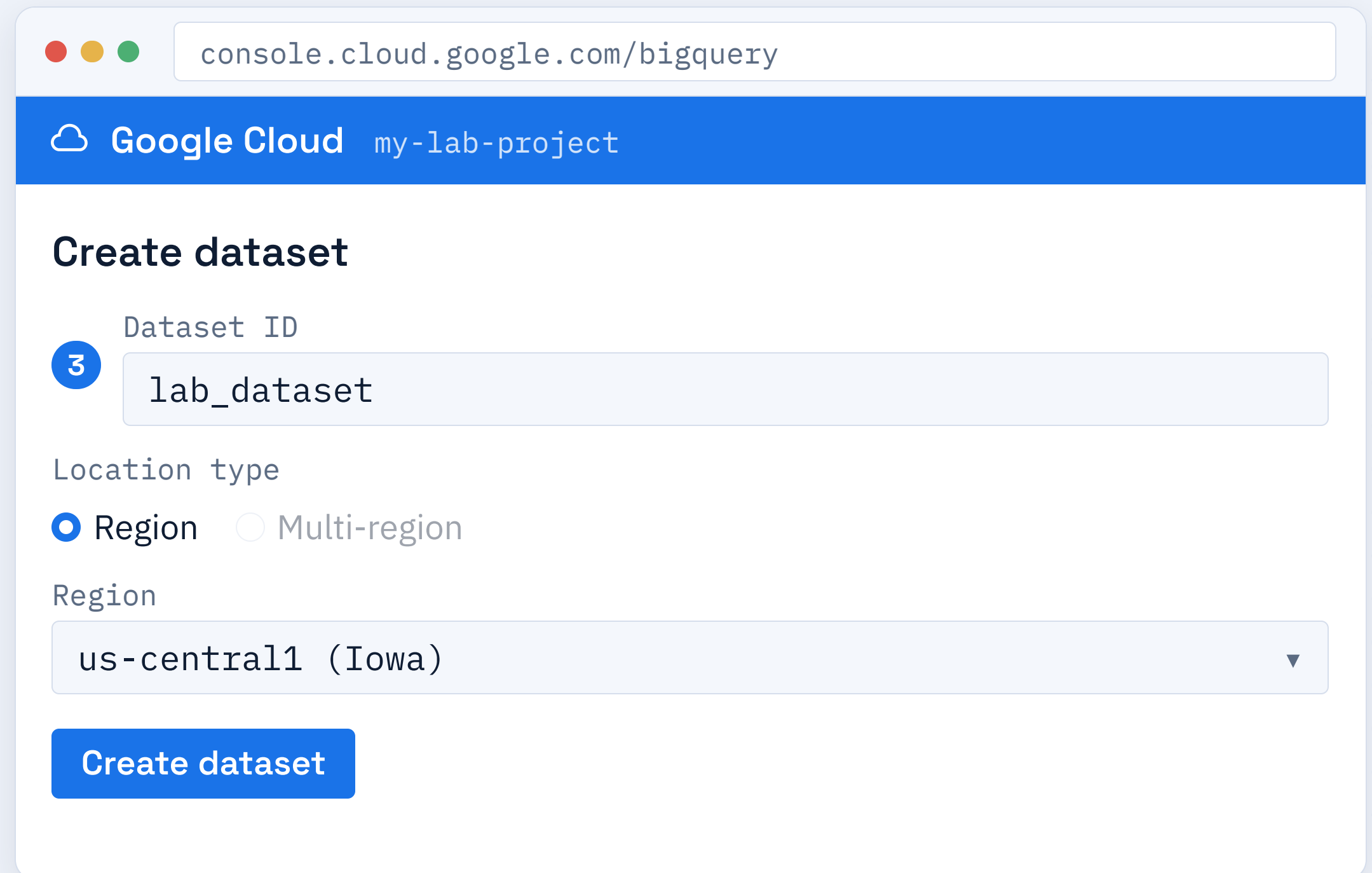
✓ Message successfully acknowledged and removed from queue.

Create a Dataset

STEPS

- 1 Open BigQuery.
- 2 Next to your project, click **Create Dataset**.
- 3 Enter a Dataset ID, e.g. `lab_dataset`.

✓ **Expected result:** the dataset appears under your project in the explorer.



The screenshot shows a web browser window with the address bar displaying `console.cloud.google.com/bigquery`. The page header is blue with the Google Cloud logo and the text "my-lab-project". The main content area is titled "Create dataset". It contains a form with the following fields:

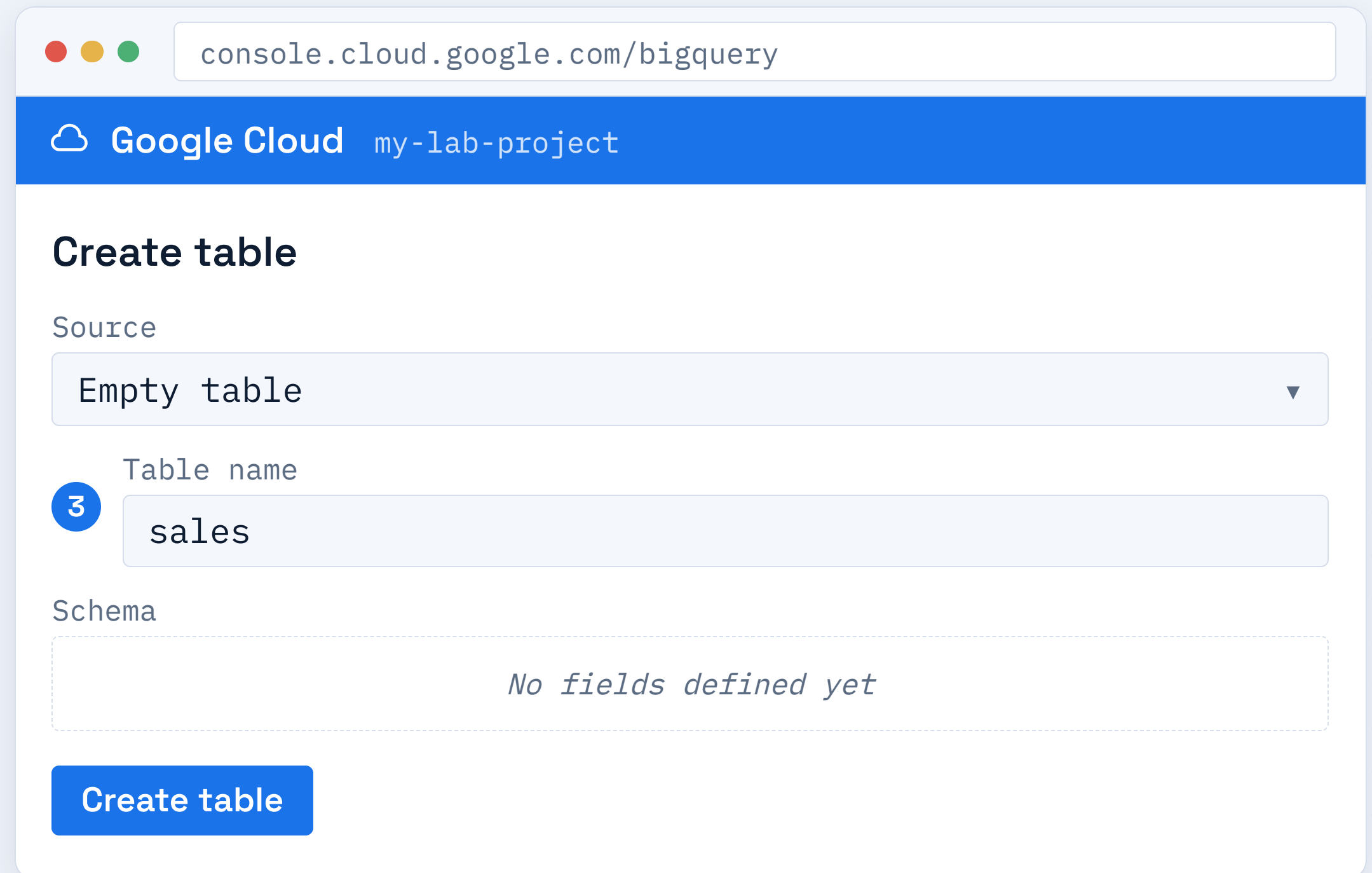
- Dataset ID:** A text input field with the value "lab_dataset".
- Location type:** Two radio buttons, "Region" (selected) and "Multi-region".
- Region:** A dropdown menu showing "us-central1 (Iowa)".
- Create dataset:** A blue button at the bottom of the form.

Create a Table

STEPS

- 1 Open the dataset `lab_dataset`.
- 2 Click **Create Table**.
- 3 Name the table, e.g. `sales`.

✓ **Expected result:** an empty table is created in the dataset.



The screenshot shows the Google Cloud BigQuery console interface. At the top, the browser address bar displays `console.cloud.google.com/bigquery`. Below the address bar, a blue header bar contains the Google Cloud logo and the text "my-lab-project". The main content area is titled "Create table". Under the "Source" label, a dropdown menu is set to "Empty table". Under the "Table name" label, a text input field contains the word "sales", with a blue circle containing the number "3" next to it. Under the "Schema" label, a dashed border box contains the text "No fields defined yet". At the bottom of the form is a blue button labeled "Create table".

Load Data from a CSV

STEPS

- 1 In **Create Table**, set Source to **Upload**.
- 2 Choose a CSV file, e.g. [sales.csv](#).
- 3 Enable **Auto-detect schema**.

✓ **Expected result:** the CSV loads and the table shows rows.

console.cloud.google.com/bigquery

Google Cloud my-lab-project

Create table

Create table from

Upload ▼

Select file

sales.csv Browse

File format

CSV ▼

☒ **Auto detect** (Schema and input parameters) 3

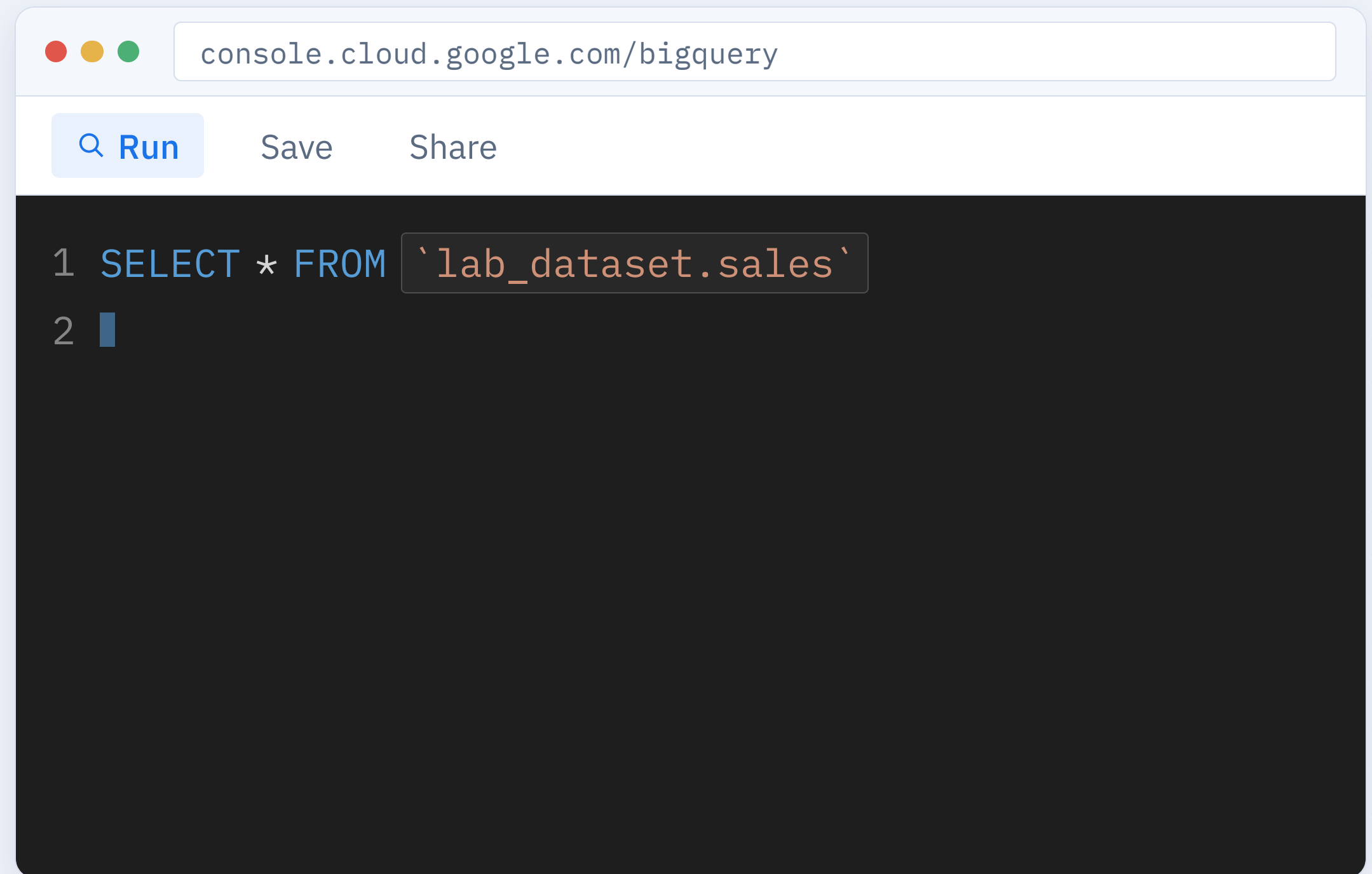
Create table

Open the Query Editor

STEPS

- 1 Select the table `sales`.
- 2 Click **Query → In new tab**.
- 3 The SQL editor opens.

✓ **Expected result:** the query editor opens referencing your table.

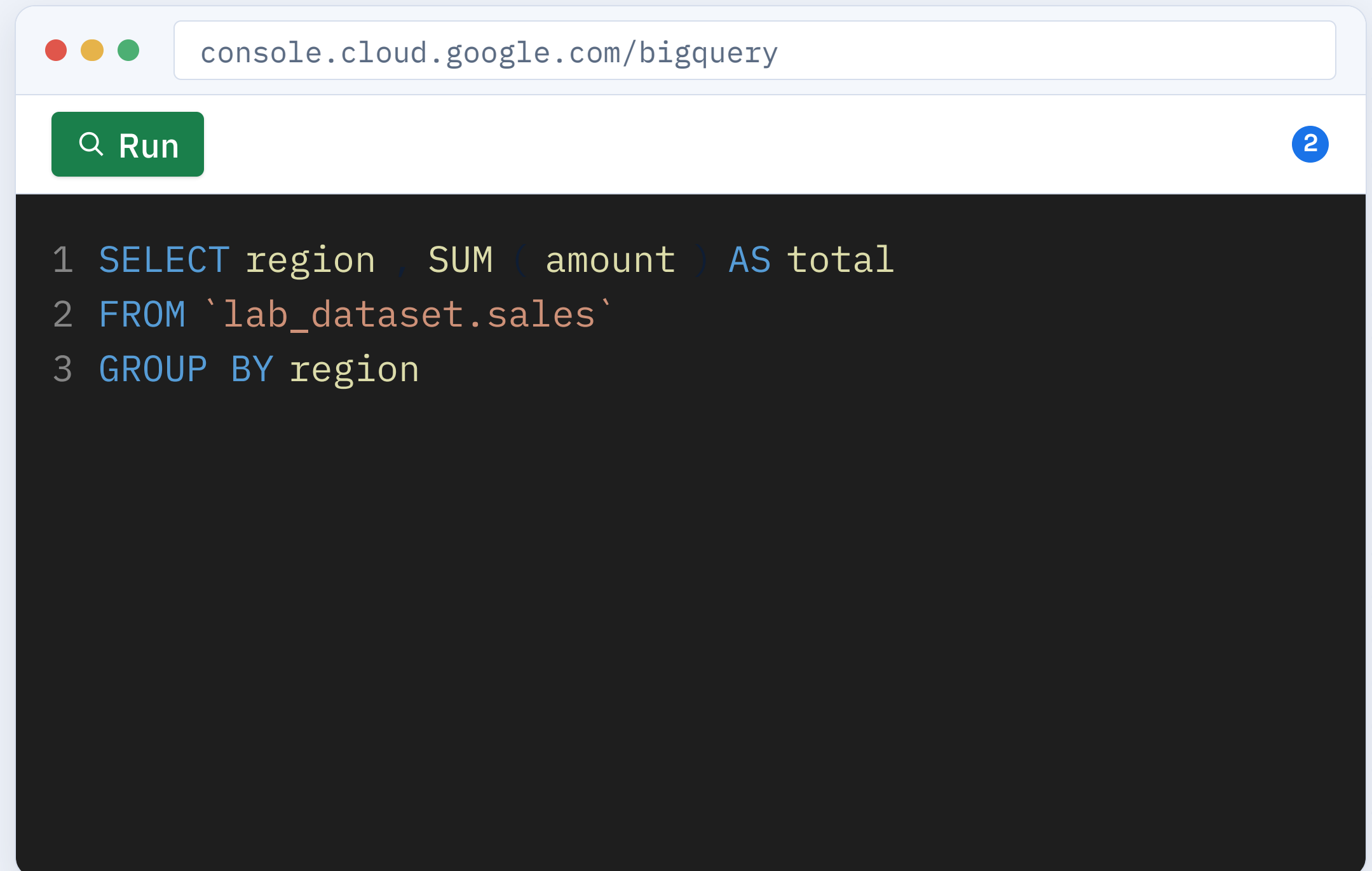


Run a SELECT Query

STEPS

- 1 Type a **SELECT** statement.
- 2 Click **Run** (the green button).
- 3 Wait for the job to finish.

✓ **Expected result:** the query runs and a results tab appears.

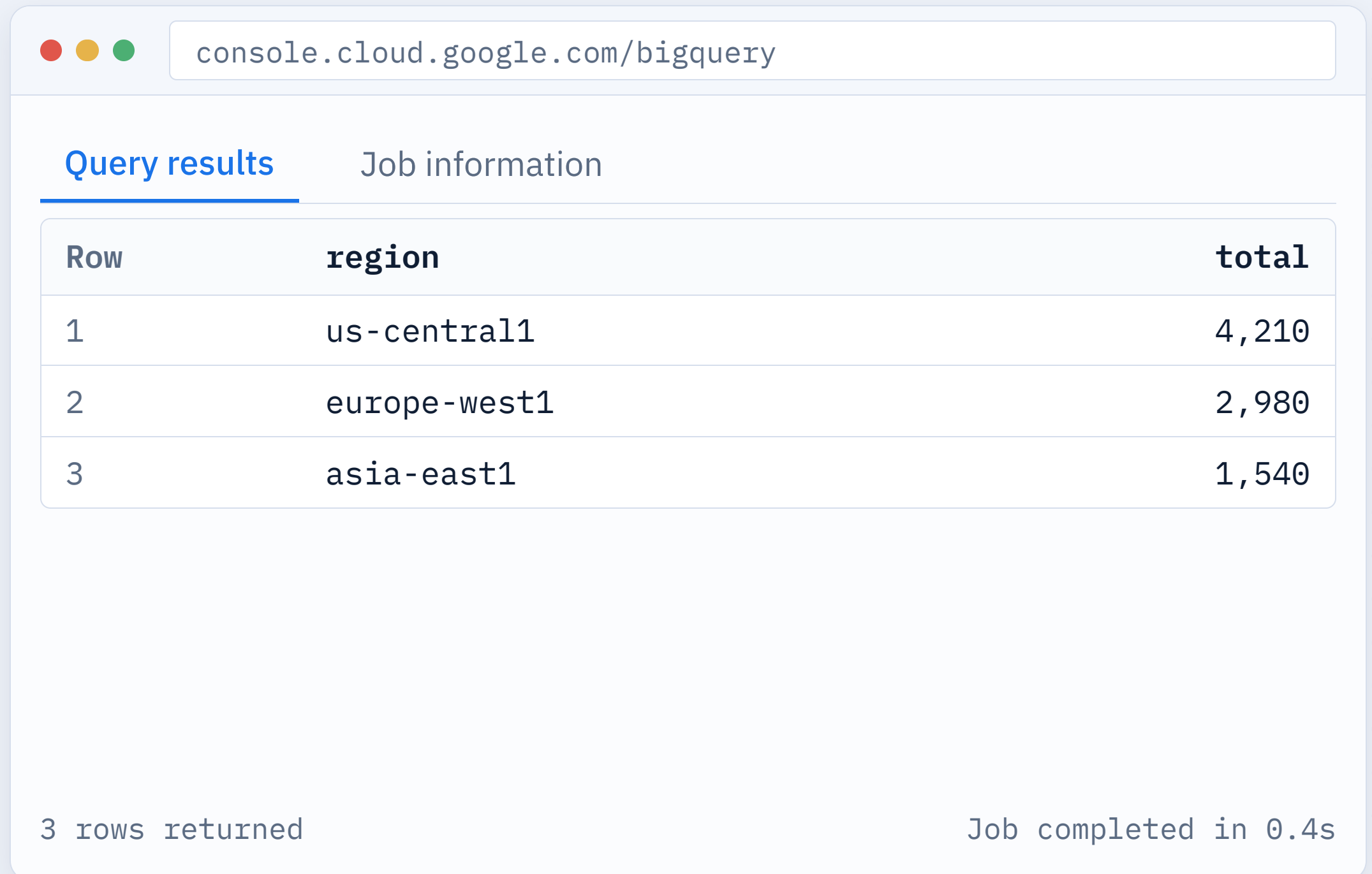


Read the Results

STEPS

- 1 Open the **Query results** tab.
- 2 Review the returned rows in the table.
- 3 Note rows scanned and job execution time.

✓ **Expected result:** the results table shows aggregated rows.



The screenshot shows the Google Cloud BigQuery console interface. At the top, there's a browser address bar with the URL 'console.cloud.google.com/bigquery'. Below the address bar, there are two tabs: 'Query results' (which is selected and underlined) and 'Job information'. The 'Query results' tab displays a table with three columns: 'Row', 'region', and 'total'. The table contains three rows of data. At the bottom of the console, there are two status messages: '3 rows returned' on the left and 'Job completed in 0.4s' on the right.

Row	region	total
1	us-central1	4,210
2	europa-west1	2,980
3	asia-east1	1,540

3 rows returned

Job completed in 0.4s

Query Pitfalls

Table not found

Wrong dataset or table name provided in the query.

Fix: use `dataset.table` and check spelling.

Syntax error

Invalid SQL structure or missing clauses.

Fix: fix the clause BigQuery points to in the editor.

Schema mismatch

CSV columns don't match the target table types.

Fix: use auto-detect or manually fix the schema definition.

Permission denied

Missing BigQuery role to access data or run jobs.

Fix: grant BigQuery Data Viewer or Editor role.

Put It All Together

Combine what you built into one small pipeline.

1

HOST

Run a small web server on your VM and reach it through the firewall rule.

2

STORE

Upload its output file to your Cloud Storage bucket.

3

ANALYZE

Load that file into BigQuery and run a query over it.

No step-by-step here — use the labs as your reference.

Map the Architecture

In the sample below, identify which service family each box belongs to.



● Compute

● Messaging

● Analytics

● Storage

Final Lab Checklist

Confirm each lab is complete before you tear down resources.

 VM created and reachable over SSH

 Bucket created with a file uploaded

 BigQuery query returned results

 Firewall rules allow HTTP and HTTPS

 Pub/Sub message published and pulled

 All resources deleted to avoid charges

5 Things You Mastered

You have built the core components of a cloud-native architecture.



Compute & Networking

Deploying VMs and securing them with VPC firewall rules.



Object Storage

Managing global buckets and unstructured data objects.



Asynchronous Messaging

Decoupling systems with Pub/Sub topics and subscriptions.



Data Analytics

Running SQL queries on massive datasets with BigQuery.



Security Best Practices

Applying least privilege and avoiding public resource exposure.